

The Symmetry-Quotient Attack on Odd-Characteristic SNOVA

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Abstract

We give new forgery attacks on the SNOVA multivariate signature scheme, targeting the odd-characteristic ($q = 19$) Version 2.3 that was proposed as a response to earlier attacks on the characteristic-two submission. Our attacks break all nine published parameter sets.

The attacks exploit a symmetry in the odd-characteristic public key. Because the public matrices are symmetric, the two ordered feature labels (ξ, η) and (η, ξ) name the same polynomial, so each key carries strictly fewer independent quadratic features than intended. We turn this collapse into a complete square system of forgery equations for every parameter set and solve it directly, keeping every verifier equation and the original rejection rule.

The resulting attack costs are $2^{132.2}$, $2^{183.8}$, and $2^{233.8}$ operations at NIST security Levels I, III, and V. These lie 10.8, 23.2, and 38.2 bits below the 143-, 207-, and 272-bit references that the parameters are meant to meet, so the scheme falls short of its target security at every level. A conservative fixed-core variant that needs no adaptive preflights still costs at most $2^{142.7}$, $2^{185.5}$, and $2^{233.8}$. Patching the scheme while keeping its architecture requires at least 45%–61% more verifier outputs.

1 Introduction

The signature schemes deployed today rest on factoring and discrete logarithms, both of which a large quantum computer would break. This is why NIST is standardizing replacements built on problems believed hard even for quantum computers. *Multivariate* schemes are one such family: their security rests on the difficulty of solving systems of quadratic equations in many variables over a finite field.

Concretely, a multivariate scheme publishes a map

$$P : \mathbb{F}_q^N \longrightarrow \mathbb{F}_q^M,$$

each coordinate a polynomial of degree at most two over the finite field $\mathbb{F}_q$. A signature is a preimage under P : to sign, the legitimate signer uses secret trapdoor structure to solve $P(x) = t$ for a hash-derived target t , and the verifier simply checks $P(x) = t$. An attacker with no trapdoor must solve the same equations directly, and pass whatever format and rejection checks the verifier applies. Counting equations against unknowns only estimates how hard this is. What actually decides it is structure: hidden dependencies among the quadratic terms can expose linear constraints, kill whole output coordinates, or split the system into smaller solvable pieces. Attacks on multivariate schemes almost always exploit such structure, and this paper is no exception.

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SNOVA is one of nine digital-signature candidates that NIST advanced to the third round of its additional post-quantum signature process in May 2026 [1, 2]. Its original construction, over a characteristic-two field, was broken through a *wedge* structure [3, 4]. Version 2.3 is the proposed fix: it moves to the odd-characteristic field $q = 19$ and makes the public base matrices symmetric [5]. We attack Version 2.3. As we now explain, the symmetry that was added to repair the scheme is the very thing that breaks it.

The weakness: a hidden duplication. Each public key is built from base matrices P_i that the scheme expands using a small matrix algebra $A = \mathbb{F}_q[S]$ of dimension d . It forms its quadratic building blocks by sandwiching each P_i between a left label ξ and a right label η , both drawn from A (here $\Sigma_\xi = I_n \otimes \xi$):

$$q_{i,\xi,\eta}(u) = u^\top \Sigma_\xi P_i \Sigma_\eta u.$$

Because P_i is now symmetric and this expression is a scalar, hence equal to its own transpose,

$$q_{i,\xi,\eta}(u) = q_{i,\eta,\xi}(u). \quad (1)$$

Swapping the two labels returns the *same* polynomial. The verifier gives these two ordered labels separate names and treats them as independent features, but they are one feature counted twice. Tallying what survives, the d^2 ordered label pairs collapse onto the symmetric square $\text{Sym}^2(A)$, leaving at most

$$\dim_{\mathbb{F}_q} \text{Sym}^2(A) = \binom{d+1}{2}$$

genuinely independent features. For the published parameters $d = \ell$, so the 16 nominal features per base matrix become 10 when $\ell = 4$, and the 4 become 3 when $\ell = 2$. This collapse happens before the public map mixes the features together, so no amount of later mixing can rebuild the directions that have already vanished.

From duplication to a forgery system. A duplicated feature by itself is only a hint; we turn it into a full forgery procedure. Beullens showed that an attacker can constrain a SNOVA signature to lie in a chosen affine subspace, which shrinks the search [6]. Working inside that framework, we use the symmetry to separate the verifier’s output coordinates into two kinds. The coordinates that the duplicated features can no longer reach turn into plain linear or consistency conditions, which are cheap to satisfy. The rest form a compact system of K quadratic equations in K unknowns—a *square* system—that carries all the remaining hardness. Every step is a public, reversible transformation that preserves the verifier’s equations, so any solution we find maps back to a genuine signature that the unmodified verifier accepts.

The complete data flow is

public verifier $\longrightarrow$ affine column relation $\longrightarrow$ symmetry quotient,

exact output split $\longrightarrow$ square residual system $\longrightarrow$ reconstructed, verifier-checked signature.

Our solver may fail and retry, but it can never cause the verifier to accept an invalid signature.

Attack costs for all nine parameter sets. For every one of the nine published parameter sets, this reduction leaves a single square system to solve, and we solve it directly. This avoids the delicate case analysis and parameter tuning that earlier cryptanalyses of SNOVA relied on. Table 1 lists the resulting costs.

Table 1: Cost of our attack on all nine Version 2.3 parameter sets, as $\log_2$ of the number of field operations. The “direct” column solves the square system uniformly; the “adaptive” column uses an optional test that can select a smaller subsystem. “Nominal reference” is the bit-security level NIST assigns each set. An implementation multiplier κ_{hom} , discussed below, is excluded.

Level	Parameters (v, o, ℓ, r)	Direct square	Adaptive	Nominal reference
I	(28, 5, 4, 4)	142.693	128.332	143
I	(48, 16, 2, 2)	132.971	132.188	143
I	(28, 4, 4, 5)	142.692	128.331	143
III	(40, 7, 4, 4)	185.537	171.702	207
III	(72, 24, 2, 2)	183.795	183.795	207
III	(38, 5, 4, 5)	185.536	171.702	207
V	(50, 9, 4, 4)	227.646	214.644	272
V	(96, 32, 2, 2)	233.844	233.844	272
V	(52, 6, 4, 6)	227.645	214.643	272

Taking the worst case at each security level, the direct attack costs $2^{142.7}$, $2^{185.5}$, and $2^{233.8}$ operations at Levels I, III, and V. The optional preflight test selects a smaller subsystem when it applies and falls back to the full square when it does not, lowering the worst-case costs to

$$\boxed{2^{132.2}, \quad 2^{183.8}, \quad 2^{233.8}}.$$

Each of these sits 10.8, 23.2, and 38.2 bits below NIST’s references of 143, 207, and 272 bits [7], so every parameter set falls short of its target security. These counts omit an implementation constant κ_{hom} ; even charging a generous $\kappa_{\text{hom}} = 2^7$, the attacks stay 3.8, 16.2, and 31.2 bits under the references.

Why the system has usable solutions. Solving a square system helps only if it has a solution that the verifier accepts and that the solver can find—one where the relevant Jacobian is nonsingular. We prove such solutions exist by controlling the system’s *rank spectrum*: how often its derivatives lose rank in random directions. Section 6 makes this precise and gives an explicit lower bound on the chance of landing on a good solution. We establish the bound two ways: under a standard idealization of SNOVA’s hash-based key expansion, and, for any fixed key, by an exact finite computation.

What we do and do not claim. The reduction itself—the symmetry quotient, the output split, rejection handling, and the final verifier check—is exact and fully public. The cost figures are operation counts, not wall-clock timings: the constant κ_{hom} absorbs implementation overheads (interpolation, inversion, factorization, memory traffic, and so on) that we track explicitly rather than hide. The attack remains exponential-time; its significance is that its exponent falls below the level NIST’s parameters are meant to guarantee.

Consequences for repair. Because the weakness is structural, patching it is not a matter of a small rejection tweak. Under the current parameter coupling, merely restoring the lost features and removing the forced square system already demands 45%–61% more verifier outputs, before any security margin is added back; removing the chosen-message square by dimension count alone

demands 97%–128% more. These are lower bounds on what a repair must cost, not recipes for a secure one. A companion paper examines the broader structural limits of such symmetric public forms [8], but nothing here depends on it.

Contributions.

1. We prove that every symmetric Version 2.3 public key loses quadratic features exactly as the symmetric square predicts, independent of any basis choice.
2. We give a reduction, valid for all nine parameter sets, that turns this loss into a square forgery system while preserving every verifier equation and correctly handling the scheme’s rejection rule.
3. We prove that this system has solutions the attacker can use, with a bound that assumes no independence between the rejection rule and the equations.
4. We give explicit solving procedures—a uniform one and a faster adaptive one—with field-arithmetic circuits and a complete cost ledger for all nine sets.
5. We prove quantitative lower bounds on the cost of any repair, and include a second, self-contained solving route (streamed XL) for readers who prefer explicit linear algebra.

Roadmap. Section 2 supplies the MQ, rank-spectrum, and symbolic-homotopy background. Section 3 states the public construction and the symmetry quotient. Section 4 gives the exact affine-column and output reductions. Section 5 constructs fresh slices, handles rejection, and explains exact target sampling. The full-domain root and complete-square theorems begin in Section 6. The low-state XL complement, implementation evidence, countermeasures, and conclusion follow afterward.

2 Background and relation to prior work

2.1 MQ systems, roots, and affine slices

An MQ system over $\mathbb{F}_q$ is a map $g : \mathbb{F}_q^N \rightarrow \mathbb{F}_q^K$ whose coordinates have total degree at most two, together with a target $z \in \mathbb{F}_q^K$. Linear and constant terms are allowed. A root is an x satisfying $g(x) = z$. Underdetermination makes roots more plentiful but does not make the system linear or automatically easy.

An affine space is a translate $a + V$ of a linear subspace. Restricting the search to an affine slice $a + L \subseteq a + V$ turns the problem into a system in $h = \dim L$ coordinates. Three events must be separated:

- (i) the target–slice pair contains a base-field root;
- (ii) it contains an accepted root at which a chosen Jacobian has full rank;
- (iii) the chosen algebraic solver recovers every such root within its stated work bound.

The first two are properties of the map and slice. The third is a solver statement. The paper proves them separately.

2.2 Polar forms, rank spectra, and collisions

For a scalar quadratic polynomial f , define the polar form

$$\beta_f(x, y) = f(x + y) - f(x) - f(y) + f(0).$$

In odd characteristic this is bilinear. For a vector-valued map g , a nonzero $b \in \mathbb{F}_q^K$ selects the scalar quadratic $b \cdot g$. Its polar rank measures how many independent directions the quadratic part couples.

For a fixed slice direction y , the vector-valued derivative

$$T_y(x) = g(x + y) - g(x) - g(y) + g(0)$$

is linear in x . The value $q^{-\text{rank } T_y}$ is exactly the fraction of its codomain-dual directions that annihilate T_y , after the appropriate normalization. Summing these quantities over nonzero y controls the second factorial moment of the number of roots in a random target-coset pair. The same spectrum controls the density of points where the restricted Jacobian loses rank.

2.3 Macaulay matrices and the low-state branch

XL multiplies the input quadratics by selected monomials and treats the resulting polynomials as rows of a Macaulay matrix. Generic XL tracks total degree. Multigraded XL tracks one degree per variable block and can be much smaller when the system has a block structure [9, 10]. A one-dimensional right kernel is the easiest extraction case, but it is not the only sound one. If row reduction supplies a commuting border basis, the resulting multiplication matrices describe a finite quotient algebra; FGLM and finite-field factorization then enumerate its roots [11, 12]. The streamed-XL section later makes the additional production-degree premise fully explicit.

2.4 Finite-cardinality symbolic homotopy

The selected-degree-free branch uses a multihomogeneous homotopy theorem. We state the finite-field form used later.

Partition $N = \sum_{j=1}^r n_j$ variables into blocks $X_1, \dots, X_r$. Let $f_1, \dots, f_N$ be nonconstant polynomials represented by a straight-line program of length L , and suppose f_i has multidegree at most $\mathbf{d}_i = (d_{i,1}, \dots, d_{i,r})$. Define

$$B = C_{\mathbf{n}}(\mathbf{d}) = [\vartheta_1^{n_1} \cdots \vartheta_r^{n_r}] \prod_{i=1}^N \left(\sum_{j=1}^r d_{i,j} \vartheta_j \right), \quad (2)$$

$$B^+ = C'_{\mathbf{n}}(\mathbf{d}') = \text{coeffsum} \left(\prod_{i=1}^N \left(\vartheta_0 + \sum_{j=1}^r d_{i,j} \vartheta_j \right) \bmod \langle \vartheta_0^2, \vartheta_1^{n_1+1}, \dots, \vartheta_r^{n_r+1} \rangle \right). \quad (3)$$

Theorem 2.1 (Finite-cardinality multihomogeneous homotopy). *Let k be a perfect finite field and put*

$$E_0 = \max \left\{ \max_j \sum_i d_{i,j}, 8B^2 \right\}.$$

If $|k| \geq E_0$, a randomized symbolic-homotopy algorithm whose separating form has a coefficient vector sampled uniformly from k^N returns a zero-dimensional parametrization containing every nonsingular root with probability at least $7/8$. On every run it returns a subset of the nonsingular roots or reports failure. Its arithmetic cost is

$$\tilde{O} \left(BB^+ \left(L + \sum_{i,j} d_{i,j} + N^2 \right) N \right)$$

operations in k . More generally, if $|k| \geq \max_j \sum_i d_{i,j}$ and $|k| > B^2$, the same algorithm succeeds with probability at least $1 - B^2/|k|$.

Proof. Safety El Din and Schost prove the same homotopy construction under a large-characteristic hypothesis [13]. The two uses of that hypothesis can be replaced by finite-cardinality arguments.

For block j , put $D_j = \sum_i d_{i,j}$ and choose distinct $\alpha_{j,0}, \dots, \alpha_{j,D_j-1} \in k$. The affine moment-curve forms

$$\kappa_{j,u}(X_j) = X_{j,1} + \alpha_{j,u}X_{j,2} + \dots + \alpha_{j,u}^{n_j-1}X_{j,n_j} + \alpha_{j,u}^{n_j}$$

have the same Vandermonde properties as the start forms in the cited proof: any n_j are independent and any $n_j + 1$ are inconsistent. The product start system therefore has exactly B simple roots and is enumerated within the same soft-linear cost. The first term in E_0 supplies enough distinct nodes.

The homotopy proof next requires a well-separating linear form. It identifies at most B^2 nonzero vectors on which the coefficient vector must not vanish. The cited proof samples from a one-parameter moment-curve family, which is why its field-size condition includes an additional factor depending on N . Here the coefficient vector is instead sampled uniformly from the full space k^N . For each fixed nonzero vector, its annihilator inside k^N is a proper k -linear subspace (possibly of codimension greater than one), so a uniform coefficient vector annihilates it with probability at most $|k|^{-1}$. Since $|k| \geq 8B^2$, a union bound gives success at least $7/8$.

The remaining Newton lifting, specialization, squarefree cleaning, and zero-dimensional parametrization use only field operations and inversion at nonsingular points. Finite fields are perfect, so the rest of the cited correctness and arithmetic analysis is unchanged. Finally, direct substitution into the original system and a Jacobian-rank test remove every spurious or singular output; hence a run returns only nonsingular roots or reports failure. $\square$

The theorem is a time theorem. It does not promise a small resident state. The final parametrization can itself be enormous, and all soft- O factors, fixed transforms, factorization, filtering, memory traffic, and implementation constants are retained later in κ_{hom} .

2.5 Relation to prior attacks

Beullens rewrote the SNOVA verifier in ordered bilinear power-pair coordinates and derived affine-column forgery attacks from rank defects in the outer map [6]. Our quotient acts one layer earlier: public-matrix symmetry identifies the power-pair polynomials before the outer map sees them.

The qualitative risk of odd-characteristic symmetrization had been noted. A 2025 presentation by the SNOVA team used an alternating-form count, warned that a wedge-style variant might apply, and called for further cryptanalysis [14]. NIST later described the symmetric odd-characteristic construction as a proposed response rather than a stable design [2]. That warning did not give the exact $\text{Sym}^2(A)$ affine-column factorization, complete residual systems, target and rejection analysis, or concrete all-nine recovery costs established here.

The characteristic-two wedge attack applies to the submitted $q = 2^k$ construction [3]; our theorem targets the symmetric $q = 19$ response. Other analyses reinterpret scalar-expanded SNOVA as UOV or extension-field UOV for key recovery [15, 16, 17, 18, 19]. Our attack is a direct forgery attack and does not recover the hidden oil subspace. The complementary generic-MQ and partitioning literature includes Hashimoto’s underdefined optimizer, Just Guess, generalized partitioning, and recent F4 engineering [20, 21, 22, 23].

2.6 Notation at a glance

Table 2: Recurring notation.

Symbol	Meaning
(v, o, ℓ, r)	Public parameter tuple: vinegar blocks, oil blocks, block height, and columns per block.
$n = v + o$	Number of matrix-valued signature blocks.
$M = or\ell$	Number of scalar verifier outputs.
$N_0 = n\ell$	Length of one stacked signature column after the common-column relation.
$A = \mathbb{F}_q[S]$	Public commutative matrix algebra; in Version 2.3, $A \cong \mathbb{F}_{q^\ell}$.
$m_1 = \lceil or/\ell \rceil$	Number of public base-form groups.
$K = m_1 \binom{\ell+1}{2}$	Maximum number of distinct quotient quadratics.
Q_R	Public matrix mapping quotient features to verifier outputs for relation R .
W_R, G_R	Left-kernel basis and public left inverse used for the exact output split.
$C_{R,v}$	Affine constraint matrix after projecting away the quadratic image.
L, h	Solver slice and its base-field dimension.
$\bar{\eta}_L^U$	Full-domain directional collision mass used by the headline root theorem.
κ_{hom}	Explicit multiplier for homotopy factors not in the unit-leading ledger.

3 Public forms and the symmetry quotient

3.1 Parameters and public forms

The public parameter tuple is (v, o, ℓ, r) . There are $n = v + o$ matrix-valued signature blocks, each with ℓ rows and r columns. Stacking one column of all blocks gives a vector of length $N_0 = n\ell$. The verifier has

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad M = or\ell, \quad N_0 = n\ell. \quad (4)$$

The ceiling is the convention used by the pinned Version 2.3 reference implementation [5].

Let $S \in \text{Mat}_\ell(\mathbb{F}_q)$ be the public symmetric matrix with irreducible characteristic polynomial. Its powers generate

$$A = \mathbb{F}_q[S], \quad \Sigma_\xi = I_n \otimes \xi \quad (\xi \in A).$$

For the public rows, $A \cong \mathbb{F}_{q^\ell}$ and $\dim_{\mathbb{F}_q} A = \ell$. The scalar-expanded base matrices $P_1, \dots, P_{m_1} \in \text{Mat}_{N_0}(\mathbb{F}_q)$ are symmetric. For $\xi, \eta \in A$, define

$$B_i^{\xi, \eta}(x, y) = x^\top \Sigma_\xi P_i \Sigma_\eta y. \quad (5)$$

Setting $x = y$ gives the quadratic power-pair feature.

Table 3: Dimensions of the nine public $q = 19$ Version 2.3 parameter rows.

Level	(v, o, ℓ, r)	m_1	M	K	N_0
I	(28, 5, 4, 4)	5	80	50	132
I	(48, 16, 2, 2)	16	64	48	128
I	(28, 4, 4, 5)	5	80	50	128
III	(40, 7, 4, 4)	7	112	70	188
III	(72, 24, 2, 2)	24	96	72	192
III	(38, 5, 4, 5)	7	100	70	172
V	(50, 9, 4, 4)	9	144	90	236
V	(96, 32, 2, 2)	32	128	96	256
V	(52, 6, 4, 6)	9	144	90	232

3.2 The basis-independent quotient

For one base form, the ordered label space is $A \otimes_{\mathbb{F}_q} A$. The following theorem identifies a universal kernel.

Theorem 3.1 (Symmetric-square factorization). *Let $A = \mathbb{F}_q[S]$ have dimension d , and assume $S^\top = S$ and $P_i^\top = P_i$. For each i , the map*

$$A \otimes_{\mathbb{F}_q} A \longrightarrow \mathbb{F}_q[u]_2^{\text{hom}}, \quad \xi \otimes \eta \longmapsto u^\top \Sigma_\xi P_i \Sigma_\eta u$$

is invariant under $\xi \otimes \eta \mapsto \eta \otimes \xi$. It factors through

$$\text{Sym}_{\mathbb{F}_q}^2(A) = (A \otimes_{\mathbb{F}_q} A) / \text{span}\{\xi \otimes \eta - \eta \otimes \xi\}.$$

Consequently the direct sum over m_1 base forms has rank at most $m_1 \binom{d+1}{2}$. After any public linear map to M outputs, its rank is at most

$$\min \left\{ M, m_1 \binom{d+1}{2} \right\}. \quad (6)$$

If the minimal polynomial of S has degree ℓ , then $d = \ell$ and the ordered feature vector is obtained by duplicating the off-diagonal coordinates of an unordered feature vector.

Proof. Every element of A is symmetric. Since the displayed value is a scalar,

$$u^\top \Sigma_\xi P_i \Sigma_\eta u = (u^\top \Sigma_\xi P_i \Sigma_\eta u)^\top = u^\top \Sigma_\eta P_i \Sigma_\xi u.$$

Thus every alternating label $\xi \otimes \eta - \eta \otimes \xi$ lies in the kernel. The symmetric square has dimension $\binom{d+1}{2}$, and a later linear map cannot enlarge its image. When powers of S form a basis, the coordinate statement is exactly the duplication of each off-diagonal unordered pair into its two ordered positions. $\square$

The theorem is an upper bound: extra public dependencies can only reduce the quadratic image further. Every concrete attack checks the exact public rank before solving. The outer map cannot repair the loss because it sees only the already-equal polynomial values. The factorization itself uses only transposition symmetry. Odd characteristic is needed later for the familiar decomposition into symmetric and alternating tensors and for the polar-form analysis. The submitted nonsymmetrized $q = 16$ construction is not covered.

3.3 Frobenius-distance normal form

The symmetric square has a canonical coordinate system that will be useful for the smaller adaptive cores. For $A = \mathbb{F}_{q^d}$ and $j > 0$, define

$$\phi_j([x \otimes y]_{\text{sym}}) = xy^{q^j} + x^{q^j}y, \quad \phi_0([x \otimes y]_{\text{sym}}) = xy.$$

If d is even and $j = d/2$, the value lies in $\mathbb{F}_{q^{d/2}}$.

Theorem 3.2 (Frobenius-distance normal form). *There is an $\mathbb{F}_q$ -linear isomorphism*

$$\text{Sym}_{\mathbb{F}_q}^2(\mathbb{F}_{q^d}) \cong \begin{cases} A^{(d+1)/2}, & d \text{ odd}, \\ A^{d/2} \oplus \mathbb{F}_{q^{d/2}}, & d \text{ even}, \end{cases}$$

given by the maps ϕ_j for the distinct Frobenius distances. In particular,

$$\text{Sym}_{\mathbb{F}_q}^2(\mathbb{F}_{q^4}) \cong \mathbb{F}_{q^4} \oplus \mathbb{F}_{q^4} \oplus \mathbb{F}_{q^2}, \quad (7)$$

$$\text{Sym}_{\mathbb{F}_q}^2(\mathbb{F}_{q^2}) \cong \mathbb{F}_{q^2} \oplus \mathbb{F}_q. \quad (8)$$

Proof. Extend scalars to an algebraic closure and identify $A \otimes_{\mathbb{F}_q} \overline{\mathbb{F}}_q$ with a product of d copies indexed by $\mathbb{Z}/d\mathbb{Z}$. Frobenius cyclically shifts the primitive idempotents. The map ϕ_0 records the d diagonal unordered pairs. For $0 < j < d/2$, the conjugates of ϕ_j record the orbit of unordered pairs $\{i, i+j\}$, an orbit of size d . If d is even and $j = d/2$, the pair is repeated after a half-turn, leaving an orbit of size $d/2$. These orbits partition all unordered pairs. The displayed map is therefore a coordinate permutation after scalar extension and hence an isomorphism over $\mathbb{F}_q$. $\square$

For $\ell = 4$, the three channel types have highest homogeneous forms of ordinary degrees 2, $q + 1 = 20$, and $q^2 + 1 = 362$ over $A = \mathbb{F}_{19^4}$. For $\ell = 2$, the complete quotient becomes two eigenblocks with bidegree families $(2, 0)$, $(1, 1)$, and $(0, 2)$ after scalar extension.

4 The exact affine-column reduction

4.1 Imposing one common column variable

Write the stacked signature columns as

$$U = [u_0 \mid \cdots \mid u_{r-1}], \quad u_j \in \mathbb{F}_q^{N_0}.$$

The attacker chooses multipliers $R_j \in A$ and offsets $v_j \in \mathbb{F}_q^{N_0}$, with $R_0 = 1$, and imposes

$$u_j = \Sigma_{R_j} u + v_j, \quad 0 \leq j < r. \quad (9)$$

This is a publicly checkable subset of possible signatures and makes no assumption about the honest signer.

Before substitution, the verifier feature vector has $N_{\text{raw}} = m_1 \ell^2 r^2$ labeled coordinates. Let E be the public outer map after the fixed permutation that reconciles the official flattening order with the block-transposed order used here. For a fixed relation R , let R_R expand the remaining ordered power-pair features into the raw feature vector, and let $F_{R,v}$ collect the offset-linear coefficients. The complete substituted verifier has the form

$$E(R_R z_{\text{ord}}(u) + F_{R,v} u) + c_{R,v} = t. \quad (10)$$

Put

$$E_R = ER_R, \quad \Lambda_{R,v} = EF_{R,v}.$$

Products of two variable parts give the quadratics, products of one variable and one offset give the linear term, and products of two offsets give the constant.

Let $z_{\text{sym}}(u)$ contain one coordinate for each unordered pair $0 \leq a \leq b < \ell$ in every base form, and let D duplicate its off-diagonal coordinates. By [Theorem 3.1](#), $z_{\text{ord}}(u) = Dz_{\text{sym}}(u)$. Define

$$Q_R = E_R D \in \mathbb{F}_q^{M \times K}, \quad K = m_1 \binom{\ell + 1}{2}. \quad (11)$$

The columns of Q_R span every output direction reachable by the quotient quadratics.

4.2 Canonical output coordinates

Choose W_R whose rows form a basis of the left kernel of Q_R , and put

$$C_{R,v} = W_R \Lambda_{R,v}. \quad (12)$$

Multiplication by W_R kills every quadratic feature and exposes ordinary affine constraints.

Lemma 4.1 (Canonical residual coordinates). *Assume $\text{rank } Q_R = K$. Choose a public left inverse $G_R \in \mathbb{F}_q^{K \times M}$ satisfying $G_R Q_R = I_K$. Then*

$$\begin{pmatrix} G_R \\ W_R \end{pmatrix} \in \mathbb{F}_q^{M \times M}$$

is invertible. Writing $b = t - c_{R,v}$, the complete substituted verifier is equivalent to

$$C_{R,v}u = W_R b, \quad g_{R,v}(u) = G_R b, \quad g_{R,v}(u) = z_{\text{sym}}(u) + G_R \Lambda_{R,v} u. \quad (13)$$

The polar map of $g_{R,v}$ is exactly that of z_{sym} . If b is uniform in $\mathbb{F}_q^M$, the two right-hand sides in (13) are independent and uniform.

Proof. If a vector $y \in \mathbb{F}_q^M$ is killed by both G_R and W_R , then $W_R y = 0$ implies $y = Q_R a$ for some $a \in \mathbb{F}_q^K$. Applying G_R gives $a = G_R Q_R a = 0$, so $y = 0$. The stacked square matrix is therefore invertible. Applying its block rows to $Q_R z_{\text{sym}}(u) + \Lambda_{R,v} u = b$ gives the two displayed systems, and invertibility gives the converse. Affine terms have zero polar. Finally, an invertible linear transformation sends a uniform vector to a uniform vector on the product of the two coordinate spaces. $\square$

Proposition 4.2 (Exact affine reduction). *Let $\rho_Q = \text{rank } Q_R$, $\lambda = \text{rank } C_{R,v}$, and $\delta = M - \rho_Q - \lambda$. Then*

$$\text{rank}[Q_R \ \Lambda_{R,v}] = \rho_Q + \lambda.$$

After row reduction, exactly δ independent affine consistency conditions remain on the target. For every target satisfying those conditions, Gaussian elimination leaves ρ_Q equations of degree at most two in $N_0 - \lambda$ variables, with exactly the same solution set as the complete substituted verifier.

If $\rho_Q = K$ and $\lambda = M - K$, every target passes the affine consistency test and the residual system consists of K equations on an affine translate of $\ker C_{R,v}$.

Proof. The row space of W_R is $\ker Q_R^\top$, so $\ker W_R = (\ker Q_R^\top)^\perp = \text{im } Q_R$. Hence W_R realizes the quotient map $\mathbb{F}_q^M \rightarrow \mathbb{F}_q^M / \text{im } Q_R$. The image of the columns of $\Lambda_{R,v}$ in this quotient has dimension $\text{rank}(W_R \Lambda_{R,v}) = \lambda$, proving the first identity.

The projected affine system is $C_{R,v}u = W_R b$. Its codomain has dimension $M - \rho_Q$ and its image has dimension λ , leaving $\delta = (M - \rho_Q) - \lambda$ target consistency equations. Whenever they hold, parameterize the affine solution space by $N_0 - \lambda$ free variables and substitute into a complementary set of ρ_Q rows. Every operation is invertible row reduction or exact affine elimination, so the solution set is unchanged. In the full-rank case, $C_{R,v}$ is surjective and $\delta = 0$. $\square$

Operational meaning. The attacker computes $Q_R, G_R, W_R, C_{R,v}$ and the canonical residual map from the public key. Unfavorable public ranks cause a change of relation, offsets, or generated key before an expensive solve. Once the ranks pass, every solver root maps back through (9) to a candidate for the full original verifier. No equation has been dropped.

4.3 A Level-I dimension walkthrough

For $(v, o, \ell, r) = (28, 5, 4, 4)$, we have $n = 33$, $N_0 = 132$, $M = 80$, $m_1 = 5$, and $K = 50$. The common-column relation leaves 80 ordered quadratic labels, but the symmetry quotient leaves only 50 distinct quadratics. In the full-rank affine case, the other 30 output directions become linear equations and the full affine residual domain has dimension $132 - 30 = 102$. The complete-square theorem later chooses a fresh ordinary 50-dimensional slice inside that domain and solves all 50 residual equations together.

5 Stable slices, fresh coefficients, rejection, and targets

5.1 An unconditional stable kernel

Generic offsets can supply the missing affine output directions, but their linear terms may destroy the block grading used by multihomogeneous solvers. Choose one vector $w \in \mathbb{F}_q^{N_0}$ and multipliers $\Gamma_0, \dots, \Gamma_{r-1} \in A$, and set

$$v_j = \Sigma_{\Gamma_j} w. \quad (14)$$

Define the offset row space

$$U_w = \text{span}_{\mathbb{F}_q} \{w^\top \Sigma_\xi P_i \Sigma_\eta : 1 \leq i \leq m_1, \xi, \eta \in A\}. \quad (15)$$

For a basis $1, S, \dots, S^{d-1}$ of A , form the complete right- A orbit

$$O_{R,v}^F = \begin{pmatrix} F_{R,v} \\ F_{R,v} \Sigma_S \\ \vdots \\ F_{R,v} \Sigma_{S^{d-1}} \end{pmatrix}, \quad \rho_F = \text{rank } O_{R,v}^F. \quad (16)$$

Theorem 5.1 (Stable-lift kernel). *Assume $S^\top = S$, $P_i^\top = P_i$, and $A \cong \mathbb{F}_{q^d}$. Under (14):*

- (i) U_w is closed under right multiplication by A and $\dim_{\mathbb{F}_q} U_w \leq m_1 d^2$;
- (ii) every row of $F_{R,v}$ and $O_{R,v}^F$ lies in U_w ;
- (iii) the row space of $O_{R,v}^F$ is an A -subspace, so $d \mid \rho_F$;

(iv)

$$H_{R,v}^F = \ker O_{R,v}^F$$

is A -linear, satisfies $F_{R,v} H_{R,v}^F = 0$, and has

$$\dim_A H_{R,v}^F = n - \rho_F/d \geq n - m_1 d. \quad (17)$$

No generic orbit-rank equality is required. An unexpected dependency only enlarges the stable space.

Proof. The spanning family in (15) has $m_1 d^2$ members. For $\zeta \in A$,

$$(w^\top \Sigma_\xi P_i \Sigma_\eta) \Sigma_\zeta = w^\top \Sigma_\xi P_i \Sigma_{\eta\zeta},$$

so U_w is right- A stable. Every offset cross term has one occurrence of w ; if it appears on the right, transpose the scalar and use symmetry and commutativity to put it on the left. Thus all rows of $F_{R,v}$ lie in U_w , and right stability places all orbit rows there as well.

Because (16) contains the complete right orbit of every row, its row space is an A -vector space, proving $d \mid \rho_F$. Its first block is $F_{R,v}$, so the kernel kills every offset-linear feature. If x lies in the kernel and $\zeta \in A$, each product $S^a \zeta$ is an $\mathbb{F}_q$ -linear combination of the chosen basis powers; therefore the same orbit rows kill $\Sigma_\zeta x$. The kernel is A -linear. Rank-nullity over A gives (17). $\square$

Since $C_{R,v} = W_R E F_{R,v}$, the stable kernel lies inside $\ker C_{R,v}$. Translation along a subspace $L \subseteq H_{R,v}^F$ changes the quotient quadratics but leaves every offset-linear feature constant.

5.2 Eigenblocks and multidegrees

Let Ω split the minimal polynomial of S , and let $e_0, \dots, e_{\ell-1}$ be the primitive idempotents of $A \otimes_{\mathbb{F}_q} \Omega$. If L is A -linear of A -dimension s , then

$$L \otimes \Omega = L_0 \oplus \dots \oplus L_{\ell-1}, \quad \dim_\Omega L_a = s.$$

The unordered power-pair features are an invertible recombination of the unordered eigenblock features

$$x^\top \Sigma_{e_a} P_i \Sigma_{e_b} x, \quad 0 \leq a \leq b < \ell.$$

They have multidegree $2\mathbf{e}_a$ when $a = b$ and $\mathbf{e}_a + \mathbf{e}_b$ when $a < b$. Translation creates lower-degree terms only in the same one or two blocks. One homogenizer per block therefore preserves these exact multidegrees. This is the grading used by the complementary streamed-XL branch.

Proof of the eigenblock statement. The power basis and the idempotent basis of $A \otimes \Omega \cong \Omega^\ell$ are related by an invertible Vandermonde matrix. Passing to the symmetric square gives an invertible change between unordered power-pair and unordered eigenblock coordinates. The idempotent e_a projects $L \otimes \Omega$ onto L_a . A diagonal feature uses one block twice; an off-diagonal feature is bilinear in two blocks. Translation and homogenization preserve the claimed supports. $\square$

5.3 Reserved-line decoupling and the structural preflight

Write the public coordinate space as an A -linear direct sum

$$V_{\text{pub}} = W \oplus V' \oplus O_{\text{pub}}, \quad \dim_A W = 1, \quad \dim_A V' = v - 1, \quad \dim_A O_{\text{pub}} = o. \quad (18)$$

Choose $0 \neq w \in W$ for (14). The relation, offset multipliers, affine matrix, stable kernel, and a canonical bounded list of candidate slices can be chosen from the outer-map data and the restrictions of the public forms with one argument in W . They do not use the internal coefficients on $V' \times V'$.

Lemma 5.2 (Reserved-line decoupling). *Under (18), $Q_R, C_{R,v}, O_{R,v}^F$, the stable kernel, and every candidate slice selected canonically from it depend only on the cross data involving W and the public outer map. In the random-XOF idealization they are independent of the internal public coefficients $P_i|_{V' \times V'}$.*

Proof. Every structured offset is in the line W . Hence every offset-linear row is a linear combination of forms $(\Sigma_\xi w)^\top P_i \Sigma_\eta$, whose left argument remains in W . The internal $V' \times V'$ block never appears, and this remains true after taking the complete right- A orbit. Kernels, canonical row-reduced bases, and candidate subspaces derived from these matrices are therefore functions of the cross data alone. $\square$

The following theorem records the random-key density of the exact structural preflight for the six $\ell = 4$ rows. Put $\rho = 14/256$, the maximum atom of one XOF byte reduced modulo 19, and let $|A| = q^d$. For the relation $R_j = S^j$, choose $\Gamma_0 = 1$ and $\Gamma_1 = 0$. Define

$$\delta_{\text{src}} = \frac{q^{m_1 \binom{d}{2}} - 1}{q - 1} \rho^{d(v-1)}, \quad (19)$$

$$\delta_{\text{proj}} = \rho^{m_1 \binom{d+1}{2}} + \left(\frac{|A|^{1+o} - 1}{|A| - 1} - 1 \right) \rho^{m_1 d^2}, \quad (20)$$

$$p_M(\rho) = \prod_{c=1}^M (1 - \rho^c), \quad p_{\text{pre}} = p_M(\rho)(1 - \delta_{\text{src}})(1 - \delta_{\text{proj}}). \quad (21)$$

Theorem 5.3 (Density of the complete structural preflight). *In the code-faithful random-XOF idealization, each of the six official $\ell = 4$ shapes satisfies*

- (i) $\text{rank } Q_R = K$ and $\text{rank } C_{R,v} = M - K$;
 - (ii) the stable kernel has trivial intersection with $W \oplus O_{\text{pub}}$;
 - (iii) its projection to V' is injective, so every selected slice has nonzero fresh projection;
- jointly with probability at least p_{pre} . For every official $\ell = 4$ shape,

$$p_{\text{pre}} > 0.9423222679.$$

All conditions are exact public rank tests on a fixed key.

Proof. The alternating part of the offset-linear source supplies the $M - K$ missing output directions. For a nonzero dual alternating tuple, the coefficient map on V' is injective after projection to the cross tensor between the reserved line and a nonzero A -line. A rank- $d(v-1)$ affine condition in fresh symbols therefore has probability at most $\rho^{d(v-1)}$; projectivizing and union bounding gives (19).

The event that the stable kernel meets the reserved line forces every symmetric restriction on $W \times W$ to vanish. An intersection with any other A -line in $W \oplus O_{\text{pub}}$ forces all $m_1 d^2$ cross symbols to vanish. Union bounding over the projective A -lines gives (20). On the complementary source event, choose $M - K$ affine columns independent modulo the K quotient columns. Conditional on the public-form data, each outer-map row misses the span of its predecessors with probability at least $1 - \rho^c$ when the remaining codimension is c . Multiplying these terms gives $p_M(\rho)$. The three coefficient regions are disjoint in the pinned byte expansion, so the probabilities multiply. Direct substitution for each of the six public dimension tuples gives the displayed uniform lower bound. $\square$

5.4 The guaranteed fresh ordinary square

Let

$$D_0 = V' \cap \ker C_{R,v}.$$

The intersection inequality gives

$$\dim_{\mathbb{F}_{19}} D_0 \geq 4(v-1) - (M-K). \quad (22)$$

For all six $\ell = 4$ rows this lower bound is at least K :

Parameters	$\dim D_0$ lower bound	K
$(28, 5, 4, 4), (28, 4, 4, 5)$	78	50
$(40, 7, 4, 4)$	114	70
$(38, 5, 4, 5)$	118	70
$(50, 9, 4, 4)$	142	90
$(52, 6, 4, 6)$	150	90

Consequently the attacker can choose, from cross data alone, an ordinary K -dimensional slice inside D_0 . Its internal $V' \times V'$ coefficients remain fresh for the spectrum theorem. This simple dimension fact is the reason the complete-square route applies to every $\ell = 4$ row, including $(38, 5, 4, 5)$; it does not require the smaller structured stable kernel to have A -dimension $K/4$.

5.5 Passing the block-symmetry rejection rule

Version 2.3 rejects a square $\ell = 4$ signature if any block is symmetric and rejects an $\ell = 2$ signature if more than $\lfloor n/4 \rfloor$ blocks are symmetric [5]. Rectangular $\ell = 4$ blocks are not subject to the square-block rule.

For a square $\ell = 4$ relation $R = (R_0, R_1, R_2, R_3)$, define the local skew map

$$\mathcal{S}_R : \mathbb{F}_q^4 \longrightarrow \bigwedge^2 \mathbb{F}_q^4, \quad \mathcal{S}_R(x)_{a,c} = (R_c x)_a - (R_a x)_c.$$

For any offsets, one signature block is symmetric exactly when $\mathcal{S}_R(x)$ equals a fixed offset-dependent vector. The public relation $R_j = S^j$ for the official matrix has rank $\mathcal{S}_R = 4$. Thus each block has at most one rejected common-column value, and the accepted density in the full common-column space is at least

$$\alpha_{4,n} = (1 - 19^{-4})^n. \quad (23)$$

The accepted-root theorem later uses only this global density; it assumes no independence from the MQ system.

For $\ell = 2$, the zero-offset relation used in the main theorem gives a block

$$\begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix},$$

which is symmetric exactly when $y = 0$. The exact accepted density is derived in (40). A deterministic alternative is also available: for the official matrix

$$S = \begin{pmatrix} 1 & 2 \\ 2 & 15 \end{pmatrix}, \quad R_1 = 9I + 10S = \begin{pmatrix} 0 & 1 \\ 1 & 7 \end{pmatrix},$$

choose offsets with $(v_1)_{b,0} - (v_0)_{b,1} = 1$ in every block. Then each block has fixed nonzero skew 1, so every root passes the rejection rule. This fixed-skew route is a useful independent check but is not needed for the selected-degree-free headline.

5.6 Exactly uniform official targets

The probability theorems use a uniform target over $\mathbb{F}_{19}^M$. The official binary-to-field conversion has a small modulo bias. The attacker removes it without changing the verifier. The pinned expansion uses each full eight-byte chunk for fifteen base-19 digits and a final partial chunk for the remaining digits. If a uniform b -bit chunk X is to supply a base-19 digits, retain the message-salt input only when

$$X < \left\lfloor \frac{2^b}{19^a} \right\rfloor 19^a. \quad (24)$$

Conditioned on this event, $X \bmod 19^a$ is uniform. Applying the test to each chunk makes the retained official target exactly uniform, and an invertible output change preserves uniformity. Exact evaluation over the nine public Version 2.3 lengths gives a minimum acceptance probability $0.1524622985\dots$, and hence at least 0.152462 . When target consistency requires more than the 128-bit salt space, the existential chosen-message attacker varies a message prefix as well as the salt. Target generation occurs before nonlinear solving and adds to, rather than multiplies, the solve cost.

5.7 The exact first and second moments

The accepted-root theorem rests on an exact collision count. We state it here so that the probability argument is self-contained.

Theorem 5.4 (Root-collision identity). *Let U be an $\mathbb{F}_q$ -space, let $C : U \rightarrow \mathbb{F}_q^c$ be surjective, let $L \subseteq \ker C$ have dimension h , and let $g : U \rightarrow \mathbb{F}_q^K$ have degree at most two. Sample independent uniform $d \in \mathbb{F}_q^c$ and $z \in \mathbb{F}_q^K$, then choose a uniform affine coset $\mathcal{A}$ of L inside $C^{-1}(d)$. Put*

$$Z = |\{x \in \mathcal{A} : g(x) = z\}|, \quad \mu = q^{h-K}.$$

For $0 \neq y \in L$, define

$$T_y(x) = g(x+y) - g(x) - g(y) + g(0)$$

and let $\kappa(y)$ be one if $-(g(y) - g(0)) \in \text{im } T_y$ and zero otherwise. Then

$$\mathbb{E}Z = \mu, \quad (25)$$

$$\mathbb{E}[Z(Z-1)] = \mu \sum_{0 \neq y \in L} \kappa(y) q^{-\text{rank } T_y} \leq \mu \sum_{0 \neq y \in L} q^{-\text{rank } T_y}. \quad (26)$$

The domain of T_y is the full space U because the experiment averages over the affine right-hand side as well as the coset.

Proof. Sampling d and then a uniform L -coset in $C^{-1}(d)$ is exactly uniform over all cosets of L in U . Every point of U therefore lies in the sampled coset with probability $q^{h-\dim U}$, and its image matches the independent target with probability q^{-K} . Summing over all points gives (25).

For an ordered pair of distinct roots in one coset, write the second point as $x+y$ with $0 \neq y \in L$. The collision condition is

$$g(x+y) = g(x) \iff T_y(x) = -(g(y) - g(0)).$$

It has $\kappa(y)q^{\dim U - \text{rank } T_y}$ solutions in x . Divide the total number of ordered collisions by the number $q^{\dim U - h}q^K$ of target-coset pairs to obtain (26). $\square$

6 Full-domain spectrum and complete-square recovery

The strengthened attack now has a deliberately simple primary theorem. After the exact symmetry quotient and affine split, every published row leaves a complete square residual system: K equations on a K -dimensional public slice. The full-domain spectrum theorem supplies accepted nonsingular roots, and finite-cardinality symbolic homotopy recovers them. No selected degree, chosen Jacobian core, family-count sweep, or exact-corank prediction appears in this direct route.

Layer	Mathematical statement	Status
Exact algebra	Public-matrix symmetry factors the ordered feature map through $\text{Sym}^2(A)$; the output split retains every verifier equation and reconstructs every candidate.	Unconditional for every symmetric public key passing the stated exact public ranks.
Root density	A full-domain directional-rank spectrum bounds both collisions and the singular locus of the restricted Jacobian.	Theorem in the code-faithful random-XOF idealization, or an exact projective-rank certificate for a fixed key.
Complete-square recovery	Restrict the full K -coordinate residual map to a public K -dimensional slice and solve all K equations together. Every candidate is checked by the original verifier.	Selected-degree-free and fixed-core-free Las Vegas theorem with no orbit enumeration.
Certified optimization	An exact determinant preflight may replace the complete square by a smaller channel core; failure invokes the complete square or a lower-work orbit fallback.	A certified branch optimization, not an assumption about the public key.

The direct theorem is the conceptual headline. The adaptive cores improve its arithmetic margins, but they are not needed to establish a theorem-backed forgery route for any of the nine rows. This separation makes the logical claim independent of the optimization machinery.

The numerical statements below are unit-leading AXN operation ledgers. They are not wall-clock timings, low-memory theorems, or complete attack circuits. They make no polynomial-time claim: the displayed multihomogeneous Bézout quantities are exponential, and κ_{hom} remains visible.

6.1 The spectrum required by the accepted-root theorem

Let U be a finite-dimensional $\mathbb{F}_q$ -space, let $C : U \rightarrow \mathbb{F}_q^c$ be surjective, let $g : U \rightarrow \mathbb{F}_q^K$ have degree at most two, and let $L \subseteq \ker C$ have dimension h . Write $g = g_2 + g_1 + g_0$ by homogeneous degree and define, for $0 \neq y \in L$,

$$T_y : U \longrightarrow \mathbb{F}_q^K, \quad T_y(x) = g(x + y) - g(x) - g(y) + g(0), \quad (27)$$

$$\bar{\eta}_L^U = \sum_{0 \neq y \in L} q^{-\text{rank } T_y}. \quad (28)$$

The superscript matters. The accepted-root experiment first samples a uniform affine right-hand side for C and then a uniform L -coset in that fiber. Its collision calculation averages over the whole domain U . Consequently the relevant derivative in (27) has domain U , not $\ker C$. Restricting T_y to $\ker C$ gives a different, generally larger quantity that is useful for fixed-fiber uniqueness but is not needed here.

Let $X \subseteq U$ be the assignments accepted by the original verifier and put $\alpha = |X|/|U|$. A trial samples independent uniform $d \in \mathbb{F}_q^c$ and $z \in \mathbb{F}_q^K$, and then a uniform affine coset of L inside $C^{-1}(d)$.

Theorem 6.1 (Accepted nonsingular root, full-domain form). *Assume $\bar{\eta}_L^U \leq \eta$ and $\alpha > \eta/(q-1)$. A trial contains an accepted root x satisfying*

$$\text{rank } D_L g(x) = h, \quad D_L g(x)[y] = T_y(x) + q_1(y),$$

with probability at least

$$p_{\text{ns}} = q^{h-K} \frac{a^2}{a + \eta}, \quad a = \alpha - \frac{\eta}{q-1}. \quad (29)$$

No independence among acceptance, singularity, the target, and the quadratic map is assumed.

Proof. Let Y count accepted roots in the sampled target-coset pair at which $D_L g$ has rank h , and let Z count all roots. For a projective direction $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$, the condition $D_L g(x)[y] = 0$ is an affine system in x whose linear part is T_y . It therefore holds on at most a $q^{-\text{rank } T_y}$ fraction of U . Union-bound over projective directions to obtain

$$\frac{|\{x : \text{rank } D_L g(x) < h\}|}{|U|} \leq \frac{1}{q-1} \sum_{0 \neq y \in L} q^{-\text{rank } T_y} \leq \frac{\eta}{q-1}.$$

Every point of U is selected by the random fiber-coset with probability $q^{h-\dim U}$ and matches the independent target with probability q^{-K} . Put $\mu = q^{h-K}$ and $a = \alpha - \eta/(q-1)$. Hence $\mathbb{E}Y \geq \mu a$. The ordered-pair collision identity gives $\mathbb{E}[Z(Z-1)] \leq \mu\eta$, and $Y(Y-1) \leq Z(Z-1)$. Therefore

$$\mathbb{E}[Y^2] = \mathbb{E}Y + \mathbb{E}[Y(Y-1)] \leq \mathbb{E}Y + \mu\eta.$$

The function $x \mapsto x^2/(x + \mu\eta)$ is increasing for $x \geq 0$. Cauchy-Schwarz therefore gives

$$\Pr[Y > 0] \geq \frac{(\mathbb{E}Y)^2}{\mathbb{E}Y + \mu\eta} \geq \mu \frac{a^2}{a + \eta},$$

which proves (29). This strictly sharpens the former denominator $1 + \eta$ whenever $a < 1$. $\square$

The following lemma improves the seeded theorem because it prices exactly the full-domain spectrum used above. Let $\rho = 14/256$ be the maximum atom of one public-expansion byte reduced modulo 19.

Lemma 6.2 (Full-domain fresh-spectrum theorem). *Assume q is odd and $A = \mathbb{F}_{q^d}$. Let L be fixed from coefficient data disjoint from a fresh symmetric-matrix region. Suppose every $0 \neq y \in L$ has nonzero projection $\bar{y}$ to a fresh A -space D , and every nonzero residual-output combination contains a nonzero coefficient $c \in \text{Sym}_{\mathbb{F}_q}^2(A)$ in at least one fresh public base form. If*

$$\dim_{\mathbb{F}_q} D = d + t,$$

then for all $0 \neq y \in L$ and $0 \neq b \in \mathbb{F}_q^K$,

$$\Pr[b \in \ker T_y^\top] \leq \rho^t. \quad (30)$$

Consequently

$$\mathbb{E}[\bar{\eta}_L^U] \leq (q^h - 1)q^{-K} + (q^h - 1)(1 - q^{-K})\rho^t, \quad (31)$$

$$\Pr[\bar{\eta}_L^U > (q^h - 1)q^{-K} + \varepsilon] \leq \frac{(q^h - 1)(1 - q^{-K})\rho^t}{\varepsilon}. \quad (32)$$

For the reserved-line construction one may take $D = V' \cong A^{v-1}$ and $t = d(v-2)$. For the zero-offset construction on the full public-vinegar summand one may take $D \cong A^v$ and $t = d(v-1)$.

Proof. Fix $y \neq 0$ and $b \neq 0$, and choose a fresh public base form on which the corresponding symmetric-square coefficient c is nonzero. If $c = \sum_{\nu} a_{\nu} \odot b_{\nu}$, the coefficient vector of the fresh symmetric matrix in the scalar derivative $b \cdot T_y(x)$ is

$$\Xi_{c,\bar{y}}(x) = \sum_{\nu} ((a_{\nu}\bar{y}) \odot (b_{\nu}x) + (b_{\nu}\bar{y}) \odot (a_{\nu}x)).$$

Its kernel is contained in the A -line $A\bar{y}$. Indeed, if $x \notin A\bar{y}$, project to the cross summand between the disjoint lines $A\bar{y}$ and Ax . After identifying both lines with A , this projection is the image of c under the symmetrization embedding. Because the characteristic is odd, that embedding is injective, so the projection is nonzero. Thus the coefficient map has rank at least $\dim_{\mathbb{F}_q} D - d = t$.

The event $b \in \ker T_y^{\top}$ forces a rank- t affine system in independent fresh symbols. Sequentially condition on nonpivot symbols and apply the maximum-atom bound to the pivots, proving (30). Finally,

$$q^{-\text{rank } T_y} = q^{-K} |\ker T_y^{\top}| = q^{-K} \left(1 + \sum_{b \neq 0} \mathbf{1}_{\{b \in \ker T_y^{\top}\}} \right).$$

Taking expectations gives $q^{-K} + (1 - q^{-K})\rho^t$. Sum over $y \neq 0$ to obtain (31). The excess above the unavoidable baseline $(q^h - 1)q^{-K}$ is nonnegative, so Markov's inequality gives (32). The two displayed choices of D have dimensions $d(v - 1)$ and dv , respectively. $\square$

We henceforth use the nearly ideal threshold

$$\boxed{\eta_{128}(h, K) = (q^h - 1)q^{-K} + 2^{-128}.} \quad (33)$$

This is not a heuristic claim of perfect rank. It is a theorem-backed allowance of only 2^{-128} above the exact full-rank baseline.

Corollary 6.3 (Numerical spectrum tails). *For the cost-driving slices, (32) with $\varepsilon = 2^{-128}$ gives the following base-two failure exponents. The entry in each cell is the weaker official shape when two shapes share a level.*

Route	Level I	Level III	Level V
$\ell = 4$ fast channel core	-138.12	-237.86	-371.14
$\ell = 4$ direct complete square	-95.64	-178.39	-294.67
$\ell = 2$ one-eigenblock core	-130.17	-263.46	-396.74
$\ell = 2$ complete two-block core	-62.21	-161.50	-260.80

Thus even the weakest headline event fails with probability below 2^{-62} . A fixed key can instead publish its exact projective rank histogram, which certifies (33) without the random-XOF idealization.

6.2 Core-complete recovery and the adaptive theorem

After scalar extension, partition the residual equations into multidegree families $\mathcal{F}_1, \dots, \mathcal{F}_r$. A count vector $\mathbf{k} = (k_1, \dots, k_r)$ chooses k_j linear combinations from family j , with $\sum_j k_j = h$. It is *core-complete* if every point at which the full restricted Jacobian has rank h has a nonzero h -row minor with a count vector in the chosen family.

Lemma 6.4 (Random projection within equation families). *Let $J_j \in E^{m_j \times h}$ contain the Jacobian rows in family j . Suppose some choice of k_j rows from each family stacks to a nonzero $h \times h$ minor. If $P_j \in E^{k_j \times m_j}$ is uniform, then*

$$\Pr \left[\det \begin{pmatrix} P_1 J_1 \\ \vdots \\ P_r J_r \end{pmatrix} = 0 \right] \leq \frac{h}{|E|}.$$

Proof. The determinant is a polynomial of total degree at most h in the entries of the P_j . It is not identically zero because coordinate-selector matrices recover the assumed minor. Apply Schwartz–Zippel. $\square$

Lemma 6.5 (Conservative well-separating-form success). *For a square multihomogeneous core with Bézout number B , represented over a finite field E large enough for the finite-cardinality start system, the well-separating-form construction has a set of at most B^2 nonzero forbidden coefficient vectors. A uniform linear form therefore satisfies every separation condition with probability at least*

$$\sigma_E(B) = 1 - \frac{B^2}{|E|}.$$

Conditional on this event, the finite-cardinality symbolic-homotopy algorithm returns a zero-dimensional parametrization containing every nonsingular core root, with one-sided error.

Proof. The finite-cardinality homotopy construction reduces failure of its well-separating element to vanishing on a collection of at most B^2 nonzero vectors over E . For each fixed nonzero vector, a uniform coefficient vector annihilates it with probability at most $|E|^{-1}$. The union bound proves the displayed probability. We deliberately retain this theorem-level B^2 bound: ordinary pairwise separation alone would involve fewer hyperplanes, but it does not by itself verify every condition required by the cited homotopy parametrization. $\square$

Theorem 6.6 (Core-complete Las Vegas recovery). *Assume (33) and let $\mathcal{P}$ be a finite core-complete family. Run every count vector in $\mathcal{P}$ with fresh within-family projections and a valid finite-cardinality homotopy field. Filter every parametrization through all residual equations, base-field descent, the signature rejection rule, affine reconstruction, and the original verifier. If $W_{\mathbf{k}}$ is the unit-leading invocation cost before root, projection, and separator success factors, then*

$$\mathbb{E}W \leq \frac{\sum_{\mathbf{k} \in \mathcal{P}} W_{\mathbf{k}}}{p_{\text{ns}} \min_{\mathbf{k}} (1 - h/|E_{\mathbf{k}}|) \min_{\mathbf{k}} \left(1 - \frac{B_{\mathbf{k}}^2}{|E_{\mathbf{k}}|} \right)}. \quad (34)$$

The algorithm is Las Vegas and assumes no selected Macaulay degree, exact corank, unique root, or predetermined nonzero Jacobian core. Its actual work is κ_{hom} times the displayed ledger.

Proof. By Theorem 6.1, a trial contains an accepted full-Jacobian root with probability at least p_{ns} . At such a root, core completeness supplies a count vector with a nonzero minor. The family-projection lemma preserves it with the displayed conditional probability, and the separating form succeeds with the second displayed conditional probability. The corresponding homotopy invocation then contains the root in its output. Running every count vector guarantees that this invocation is present. Summing invocation costs and inverting the common success lower bound proves (34). Complete public filtering prevents an invalid output. $\square$

Lemma 6.7 (The 55 four-block patterns form 16 Frobenius orbits). *Index the ten unordered eigenblock families by $\{\{a, b\} : 0 \leq a \leq b < 4\}$ and cyclically add one to both indices. The nonnegative deficit vectors of total weight two form 55 vectors and exactly 16 cyclic orbits.*

Proof. There are $\binom{11}{2} = 55$ weight-two vectors. A quarter-turn fixes one, a half-turn fixes seven, and a three-quarter-turn fixes one. Burnside gives $(55 + 1 + 7 + 1)/4 = 16$. $\square$

A *fast-core preflight* is an exact public test proving that the highest homogeneous part of the selected core’s Jacobian determinant is not the zero polynomial. In the applications below, evaluating that highest part at one fixed public point suffices.

Theorem 6.8 (Adaptive certified-core fallback). *Fix a public construction for which a common structural and spectrum event $\mathcal{S}$ makes a core-complete fallback valid. Suppose:*

1. *on passing an exact public fast-core preflight, the fast route has expected unit-leading work at most W_f ;*
2. *on failing it, the core-complete fallback has expected work at most W_F ;*
3. *under the random-XOF key distribution, the preflight fails with probability at most δ , while $\Pr[\mathcal{S}^c] \leq \varepsilon$.*

For every fixed key in $\mathcal{S}$, the algorithm “test, then choose fast or fallback” is selected-degree-free and fixed-core-free. Under the random-XOF distribution, its conditional expected work obeys

$$\mathbb{E}[W \mid \mathcal{S}] \leq W_f + \min\left\{1, \frac{\delta}{1 - \varepsilon}\right\} (W_F - W_f)_+. \quad (35)$$

If an independent exact structural preflight succeeds with probability p_{str} , the random-key-normalized ledger divides the right side by $p_{\text{str}}(1 - \varepsilon)$. Without independence one may instead use any proved lower bound on $\Pr[\mathcal{S}]$.

Proof. On a fixed key, passing the preflight is a certificate, not a probabilistic premise. If it passes, the fast theorem applies; if it fails, core completeness applies. Thus no preferred core is assumed. For the expected work, let F be preflight failure. Then

$$\Pr[F \mid \mathcal{S}] \leq \frac{\Pr[F]}{\Pr[\mathcal{S}]} \leq \min\{1, \delta/(1 - \varepsilon)\}.$$

Conditioning on the two branches gives $(1 - q)W_f + qW_F = W_f + q(W_F - W_f)$ when $W_F \geq W_f$; if $W_F < W_f$, always using the fallback only helps, yielding the positive-part form. The last statement is the ordinary normalization by the density of generated keys that reach the solver. $\square$

7 Constructive arithmetic and the homotopy ledger

The primary ledger below charges complete multiplication circuits in the fixed towers $\mathbb{F}_{192}$ and $\mathbb{F}_{194}$. For a larger homotopy extension it charges the $2r - 1$ pointwise tower-field multiplications. Evaluation, interpolation, extension-basis transforms, additions, inversions, factorization, control, memory, and all remaining hidden constants stay in κ_{hom} .

Theorem 7.1 (Constructive 150-gate scalar multiplication). *The circuit model assigns unit cost to two-input AND, XOR, and XNOR gates, with free constants, wires, and fan-out. For canonical*

five-bit representatives, the following complete circuit sizes are constructive:

	addition	negation	subtraction	multiplication
$\mathbb{F}_{19}$	42	16	58	150
$\mathbb{F}_{19^2}$	84	32	116	692
$\mathbb{F}_{19^4}$	168	64	232	2628.

The 150-gate scalar netlist is correct on all $19^2 = 361$ canonical input pairs. The tower representations are

$$\mathbb{F}_{19^2} = \mathbb{F}_{19}[u]/(u^2 + 1), \quad \mathbb{F}_{19^4} = \mathbb{F}_{19^2}[v]/(v^2 - (1 + u)).$$

They are exactly isomorphic to the manuscript basis $\mathbb{F}_{19}[t]/(t^4 - t - 1)$ via

$$u = 1 + 2t + 15t^2 + 5t^3, \quad v = 4 + 13t + 13t^2 + t^3,$$

and the basis $(1, u, v, uv)$ has determinant $16 \neq 0$ modulo 19.

Proof. For a canonical scalar $x = (x_0, \dots, x_4)$, the circuit first converts to balanced signed magnitude. On the valid inputs $0, \dots, 18$, set

$$a = x_1x_3, \quad b = x_2x_3, \quad s = a \oplus b \oplus ax_2 \oplus x_4,$$

$$(m_0, m_1, m_2, m_3) = (x_0 \oplus s, x_1 \oplus s, x_2, x_3 \oplus b).$$

A direct truth-table check gives $s = 0, m = x$ for $x \leq 9$ and $s = 1, m = 19 - x$ for $10 \leq x \leq 18$. The two decoders use 18 gates. An exact four-by-four magnitude product together with the output sign reaches gate 80. Since the magnitude product is at most 81, write it as $\ell + 32h$ with $h \in \{0, 1, 2\}$, replace $32h$ by $13h \bmod 19$, fold once, and reduce the reachable five-bit values; this reaches gate 130. A 20-gate signed recoding maps r to either r or $-r \bmod 19$. The released netlist contains all 150 gates and an independent evaluator checks every canonical pair and canonical output.

For $u^2 = -1$, Karatsuba uses three scalar products, two input sums, and shared output negations, giving

$$5 \cdot 42 + 2 \cdot 16 + 3 \cdot 150 = 692.$$

At the second tower level, three $\mathbb{F}_{19^2}$ products cost $3 \cdot 692$. The two input sums cost $2 \cdot 84$, multiplication by $1 + u$ costs one scalar addition and subtraction, the low output costs one $\mathbb{F}_{19^2}$ addition, and the high output forms $p_0 + p_1$ once and subtracts it from p_2 . Hence

$$2 \cdot 84 + 3 \cdot 692 + (42 + 58) + 84 + (84 + 116) = 2628.$$

Direct polynomial arithmetic verifies $u^2 = -1$, $v^2 = 1 + u$, the displayed coordinates, and the basis determinant. $\square$

For a core over $A \in \{\mathbb{F}_{19^2}, \mathbb{F}_{19^4}\}$ with Bézout number B and start-node requirement D , let g_A be 692 or 2628 and define

$$\Gamma_A(B, D) = \min_{\substack{r \geq 1: \\ |A|^r \geq D, \\ 2r-1 \leq |A|}} \frac{(2r-1)g_A}{1 - B^2/|A|^r}. \quad (36)$$

Evaluation/interpolation at $2r - 1$ base-field points gives the number of charged pointwise multiplications in the numerator; the fixed transforms themselves remain in κ_{hom} . Lemma 6.5 gives the denominator. The numerical artifact exhausts all admissible r rather than fixing a blanket $8/7$ separator loss. For the legacy sixteen-orbit artifact, whose per-orbit data are retained unchanged, we conservatively retain its original $8/7$ reserve and only replace the complete $\mathbb{F}_{19^4}$ multiplication cost.

8 The direct complete-square theorem

The following specialization is the main conceptual simplification. It uses all residual equations on a square slice, so no Jacobian-core choice or family-count enumeration remains.

Theorem 8.1 (Direct complete-square recovery). *Retain the setting of Theorem 6.1 and suppose $h = K$. Restrict the complete residual map g to the public affine K -dimensional slice and solve all K equations together. On the event $\bar{\eta}_L^U \leq \eta$, a trial contains an accepted root at which the Jacobian of this very system is nonsingular with probability at least*

$$p_{\square} = \frac{a^2}{a + \eta}, \quad a = \alpha - \frac{\eta}{q - 1}.$$

If every equation has ordinary degree at most two, then

$$B_{\square} = 2^K, \quad B_{\square}^+ = 2^K \left(1 + \frac{K}{2}\right).$$

Let

$$D_K = \binom{K+1}{2}, \quad L_{\square} = D_K + 2K(D_K + K + 1), \quad H_{\square} = L_{\square} + 2K + K^2.$$

Over any validated finite-cardinality homotopy field, the unit-leading work is

$$W_{\square} = \frac{\Gamma_A(2^K, 2K)}{p_{\square}} 2^{2K} \left(1 + \frac{K}{2}\right) H_{\square} K. \quad (37)$$

The algorithm is Las Vegas after complete public filtering and assumes no selected Macaulay degree, exact corank, unique root, preferred core, or orbit sweep.

Proof. With $h = K$, Theorem 6.1 applies directly to the Jacobian of the complete residual system; no projection to a subcore is made. The ordinary total-degree Bézout number of K quadratics in K variables is 2^K . In the companion homotopy coefficient, selecting the auxiliary term from no factor contributes 2^K , while selecting it from one of the K degree-two factors contributes 2^{K-1} ; hence $B_{\square}^+ = 2^K + K2^{K-1}$. The displayed straight-line program first forms all D_K quadratic monomials and then evaluates the K affine quadratics, charging a coefficient multiplication and addition for every possible term. The homotopy theorem and Lemma 6.5 give (37). Every point returned by the parametrization is checked against the full residual system, affine reconstruction, rejection rule, and original verifier. $\square$

Corollary 8.2 (Direct square attacks for all six $\ell = 4$ rows). *For every official $\ell = 4$ shape, any public key passing the exact quotient and affine-rank preflights leaves a fresh ordinary K -dimensional slice in the public-vinegar region. Applying Theorem 8.1, embedding the $\mathbb{F}_{19}$ coefficients into $\mathbb{F}_{19^2}$, and charging the complete 692-gate multiplication circuit gives normalized exponents*

142.69267,	185.53659,	227.64611
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on the worse shape at Levels I, III, and V. The corresponding nominal headroom below 143, 207, 272 is

0.30733,	21.46341,	44.35389
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bits. Thus the six $\ell = 4$ unit-leading ledgers lie nominally below their numerical references even without a fast-core preflight or the $55 \rightarrow 16$ orbit reduction. The 0.30733-bit Level-I gap is especially sensitive to the explicit multiplier κ_{hom} and is not an implementation or security margin.

Proof. The manuscript's fresh-slice inequality gives $\dim(V' \cap \ker C_{R,v}) \geq K$ for all six rows, so a cross-data-selected ordinary K -slice exists. The full-domain spectrum theorem applies with $t = 4(v - 2)$ and the threshold $\eta_{128}(K, K)$. Substitute $K = 50, 70, 90$ in (37). Exact enumeration of the separator extension degree selects $r = 13, 17, 22$ over $\mathbb{F}_{192}$; the machine-readable ledger performs the structural normalization and charges the exact square-row acceptance density. $\square$

9 The zero-offset $\ell = 2$ theorem

For the three official $\ell = r = 2$ rows put

$$A = \mathbb{F}_{192}, \quad m = m_1 = o, \quad M = 4m, \quad K = 3m.$$

Use the zero-offset column relation

$$u_0 = u, \quad u_1 = 0. \quad (38)$$

Proposition 9.1 (Exact zero-offset target split). *Let Q_R be the quotient-output matrix for (38), and assume the exact public preflight gives $\text{rank } Q_R = K$. Choose a left inverse G_R and a full-row-rank matrix W_R spanning the left kernel of Q_R . The complete substituted verifier is equivalent to*

$$W_R t = 0, \quad z_{\text{sym}}(u) = G_R t. \quad (39)$$

For a uniform official target, consistency has probability $19^{-(M-K)} = 19^{-m}$, and conditioned on it the residual target is uniform in $\mathbb{F}_{19}^K$. Chosen-message and salt inputs can therefore be filtered before solving. If one deliberately charges 2^{32} AXN operations per retained exactly uniform target, the additive target-generation exponents are

$$16 \log_2 19 + 32 = 99.96684, \quad 24 \log_2 19 + 32 = 133.95026, \quad 32 \log_2 19 + 32 = 167.93368.$$

Proof. Zero offsets make every affine and constant contribution vanish, so the substituted verifier is $Q_R z_{\text{sym}}(u) = t$. Applying the invertible output change with block rows G_R, W_R gives (39). An invertible linear transformation preserves uniformity and separates the quotient and consistency coordinates. Target generation occurs before the nonlinear solve, so its work adds to rather than multiplies the solve ledger. $\square$

In one block, (38) has matrix $\begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix}$ and is symmetric exactly when $y = 0$. The $n = v + o$ block conditions use disjoint coordinates. Thus the exact density accepted by the Version 2.3 rule is

$$\alpha_n = 19^{-n} \sum_{j=0}^{\lfloor n/4 \rfloor} \binom{n}{j} 18^{n-j}. \quad (40)$$

Choose all solver slices inside the full public-vinegar summand A^v , independently of its fresh internal symmetric coefficients. By Lemma 6.2, the fresh exponent is $t = 2(v - 1)$.

9.1 The complete two-block core

Take an A -linear slice of dimension $s = 3m/2$. Its base-field dimension is $h = 2s = K$, and the complete quotient system has m equations in each bidegree family

$$(2, 0), \quad (1, 1), \quad (0, 2).$$

It is therefore square and core-complete without selecting or discarding any residual equation.

Theorem 9.2 (Complete two-block $\ell = 2$ forger). *On the event $\bar{\eta}_L^U \leq \eta_{128}(K, K)$, a consistent target-coset trial contains an accepted full-Jacobian root with probability at least*

$$p_{\text{comp}}^{(2)} = \frac{a_2^2}{a_2 + \eta_{128}(K, K)}, \quad a_2 = \alpha_n - \frac{\eta_{128}(K, K)}{18}. \quad (41)$$

The exact multihomogeneous quantities are

$$B_m^{(2)} = 2^{2m} \binom{m}{m/2}, \quad (42)$$

$$(B_m^{(2)})^+ = B_m^{(2)} \left(1 + 2m + \frac{m^2}{m+2} \right). \quad (43)$$

Finite-cardinality homotopy over $\mathbb{F}_{192}$, Frobenius descent, and complete public filtering form a selected-degree-free, fixed-core-free Las Vegas forger. The normalized primary exponents, including the additive target filter and the 2^{-128} spectrum normalization, are

$$132.97067, \quad 183.79463, \quad 233.84399$$

for Levels I, III, and V.

Proof. Apply Theorem 6.1 with $h = K$ and the exact accepted density (40). Over the splitting field, each base form supplies one equation in each of the two diagonal families and the cross family. The Bézout number is

$$[t_0^s t_1^s](2t_0)^m(t_0 + t_1)^m(2t_1)^m.$$

The cross factor must contribute $m/2$ powers to each block, proving (42). In the companion homotopy coefficient, the term with no auxiliary variable contributes B . Deleting a diagonal factor contributes $2^{2m-1}(\binom{m}{m/2} + \binom{m}{m/2+1})$, while deleting a cross factor contributes B . Summing and using $\binom{m}{m/2+1}/\binom{m}{m/2} = m/(m+2)$ proves (43). The exact separator factor is (36); every later operation is an exact filter. $\square$

Put

$$\begin{aligned} D_s &= \binom{s+1}{2}, & C_s &= (s+1)^2, \\ L_{\text{comp}} &= 2D_s + s^2 + 4m(D_s + s + 1) + 2mC_s, \\ H_{\text{comp}} &= L_{\text{comp}} + 6m + K^2. \end{aligned} \quad (44)$$

The unit-leading work on a good spectrum event is

$$W_{\text{comp}}^{(2)} = \frac{\Gamma_{\mathbb{F}_{192}}(B_m^{(2)}, 2s)}{p_{\text{comp}}^{(2)}} B_m^{(2)} (B_m^{(2)})^+ H_{\text{comp}} K. \quad (45)$$

Divide by the exact spectrum-event density and add the target-generation ledger.

9.2 The faster Level-I core and its exact fallback

For $s \leq m$, select s diagonal equations and solve one eigenblock of A^s . A descended full root reconstructs the other block by Frobenius. The core is a square system of s ordinary quadratics, with

$$B_s = 2^s, \quad B_s^+ = 2^s(1 + s/2).$$

At a fixed nonzero public point, the fresh leading row of each selected public form is surjective onto $(A^s)^\vee$. Sequential exposure therefore gives

$$p_{\text{Jac}}^{(2)}(s) \geq \prod_{j=1}^s (1 - \rho^{2j}) > 0.9970003329. \quad (46)$$

A nonzero leading determinant has degree at most s over A , so its singular point density is at most $s/19^2$. With $\eta_s = \eta_{128}(2s, 3m)$, a target-coset trial has an accepted full root nonsingular for this core with probability at least

$$p_s = 19^{2s-3m} \max \left\{ a_s - \eta_s, \frac{a_s^2}{a_s + \eta_s} \right\}. \quad (47)$$

Here $a_s = \alpha_n - s/19^2$. These are respectively the singleton and second-moment lower bounds after subtracting the core-singular locus.

At Level I take $s = m = 16$. Run the exact leading-determinant preflight. If it passes, solve the one-eigenblock core; if it fails, solve the complete two-block system of [Theorem 9.2](#). By [Theorem 6.8](#), this is fixed-core-free by construction and has normalized exponent

$$\boxed{132.18807}. \quad (48)$$

The diagonal-only and complete-only exponents in the strengthened ledger are 132.18928 and 132.97067. The adaptive value is slightly smaller than the first rounded figure because the common spectrum normalization is charged once and the exact branch mixture is performed before taking logarithms. At Levels III and V, the complete core is already faster.

10 Adaptive $\ell = 4$ recovery

Let $A = \mathbb{F}_{19^4}$ and let $L \cong A^s$ be the public slice. Under the Frobenius-distance normal form, each public base form supplies an A -valued degree-two channel and an A -valued degree-20 channel. Write their highest homogeneous parts as

$$F_{i,0}^{\text{top}}(x) = C_{i,0}(x, x), \quad F_{i,1}^{\text{top}}(x) = C_{i,1}(x, x^{19}).$$

Lemma 10.1 (Fresh channel decomposition and row surjectivity). *Restriction of a fresh symmetric $\mathbb{F}_{19}$ -bilinear form to the injected public-vinegar copy of L , followed by the Frobenius-distance transform, is surjective onto*

$$\text{Sym}_A^2(L^\vee) \oplus \text{Bil}_A(L, L^{(19)}; A).$$

For every fixed $0 \neq x_\star \in L$, the map from one public form's fresh coefficients to the pair of leading Jacobian rows

$$(DF_{i,0}^{\text{top}}(x_\star), DF_{i,1}^{\text{top}}(x_\star)) \in L^\vee \oplus L^\vee$$

is surjective over A .

Proof. After extending scalars, the four diagonal blocks of a symmetric base form comprise one Frobenius orbit and descend to an arbitrary symmetric A -bilinear form; the four adjacent blocks form a second orbit and descend to an arbitrary bilinear form between L and its Frobenius twist. The orbit coefficient sets are disjoint, so the two choices are independent. Formal differentiation gives

$$DF_{i,0}^{\text{top}}(x_\star)[h] = 2C_{i,0}(x_\star, h), \quad DF_{i,1}^{\text{top}}(x_\star)[h] = C_{i,1}(h, x_\star^{19}).$$

Evaluation of arbitrary bilinear forms is surjective. For the symmetric case, given $\lambda \in L^\vee$, choose $\phi(x_\star) = 1$ and use

$$\frac{1}{2}(\phi \otimes \lambda + \lambda \otimes \phi - \lambda(x_\star)\phi \otimes \phi).$$

□

Choose a whole coordinates of the degree-two channel and b whole coordinates of the degree-20 channel, with $a + b = s$. The determinant preflight is exact on the public key. For its random-XOF density we use the following conservative bound, which does not assume that an invertible linear change preserves independence under the slightly biased modulo-19 distribution.

Lemma 10.2 (Projective Jacobian-preflight density). *At the fixed public preflight point, let $J \in A^{s \times s}$ be the selected leading-Jacobian matrix, where $A = \mathbb{F}_{19^4}$. Then*

$$\Pr[\det J = 0] \leq \frac{|A|^s - 1}{|A| - 1} \rho^{4s}. \quad (49)$$

For the cost-minimizing values $s = 10, 14, 18$, the failure bounds are at most $3.55350 \cdot 10^{-5}$, $6.56029 \cdot 10^{-5}$, and $1.21113 \cdot 10^{-4}$, respectively.

Proof. If J is singular, some projective direction $[y] \in \mathbb{P}^{s-1}(A)$ lies in its right kernel. Fix such a direction. The joint channel-surjectivity statement above implies that forcing every selected leading row to annihilate y imposes s independent equations over A , hence $4s$ independent affine equations over $\mathbb{F}_{19}$, on fresh public-expansion symbols. The maximum-atom bound gives probability at most ρ^{4s} . There are $(|A|^s - 1)/(|A| - 1)$ projective directions. Union-bound over them. This bound is weaker than a sequential full-rank product would be under a uniform field distribution, but it is valid for the actual biased product distribution used in the random-XOF model. □

Conditional on passing the preflight, put

$$a_{a,b} = \alpha - \frac{a + 19b}{19^4}.$$

The sharpened core-root probability is

$$p_{a,b,s} = 19^{4s-K} \frac{a_{a,b}^2}{a_{a,b} + \eta_{128}(4s, K)}. \quad (50)$$

Its exact one-block Bézout data are

$$B = 2^a 20^b, \quad B^+ = B(1 + a/2 + b/20).$$

With

$$L_{a,b,s} = \mathbf{1}_{a>0} \left[\binom{s+1}{2} + 2a \left(\binom{s+1}{2} + s + 1 \right) \right]$$

$$+ \mathbf{1}_{b>0} \left[6s + s^2 + 2b(s^2 + 2s + 1) \right],$$

the unit-leading good-key work is

$$\frac{\Gamma_{\mathbb{F}_{19^4}}(B, 2a + 20b)}{p_{a,b,s}} BB^+(L_{a,b,s} + 2a + 20b + s^2)s. \quad (51)$$

Exhaustive enumeration of all 35,63,99 admissible profiles gives

$$(s; a, b) = (10; 5, 5), \quad (14; 7, 7), \quad (18; 9, 9)$$

at Levels I, III, V. On the worse shape, the fixed-core normalized exponents are 126.346, 167.124, and 207.359.

For the simplest fixed-core-free algorithm, a failed preflight invokes the direct ordinary square of [Corollary 8.2](#). Applying [Theorem 6.8](#) gives:

Level	parameters	fast profile	fast-branch ledger	adaptive ledger	nominal gap
I	(28, 5, 4, 4)	(10; 5, 5)	126.346	128.332	14.668
I	(28, 4, 4, 5)	(10; 5, 5)	126.346	128.331	14.669
III	(40, 7, 4, 4)	(14; 7, 7)	167.124	171.702	35.298
III	(38, 5, 4, 5)	(14; 7, 7)	167.124	171.702	35.298
V	(50, 9, 4, 4)	(18; 9, 9)	207.359	214.644	57.356
V	(52, 6, 4, 6)	(18; 9, 9)	207.358	214.643	57.357

This table needs no family-count sweep. If arithmetic rather than exposition is the priority, the existing sixteen-orbit fallback lowers the adaptive exponents to 126.405, 167.289, and 207.844 on the worse shape. It is now an optional optimization rather than part of the logical headline.

The smaller-output profiles from the same exhaustive search remain useful: (9; 5, 4), (10; 7, 3), and (11; 9, 2) have normalized primary exponents 133.622, 197.672, and 261.642 on the worse shape and packed final- parametrization ceilings approximately 311 MiB, 68.4 MiB, and 14.9 MiB. They are not the headline because their purpose is output volume rather than minimum arithmetic.

11 All-nine consequences

Theorem 11.1 (One complete-square theorem for every published row). *For every published $q = 19$ Version 2.3 parameter shape, a public key passing the stated exact rank preflights has a theorem-backed Las Vegas forgery route obtained by solving a complete square residual system. It uses no selected Macaulay degree, exact-corank prediction, unique-root assumption, predetermined Jacobian core, family-count pattern, or orbit enumeration. The complete-multiplier-circuit ledgers are:*

Level	parameters	complete square	ledger exponent	nominal gap
I	(28, 5, 4, 4)	ordinary 50×50 square	142.693	0.307
I	(48, 16, 2, 2)	complete two-block square	132.971	10.029
I	(28, 4, 4, 5)	ordinary 50×50 square	142.693	0.307
III	(40, 7, 4, 4)	ordinary 70×70 square	185.537	21.463
III	(72, 24, 2, 2)	complete two-block square	183.795	23.205
III	(38, 5, 4, 5)	ordinary 70×70 square	185.536	21.464
V	(50, 9, 4, 4)	ordinary 90×90 square	227.646	44.354
V	(96, 32, 2, 2)	complete two-block square	233.844	38.156
V	(52, 6, 4, 6)	ordinary 90×90 square	227.645	44.355

The worst rows at Levels I, III, and V have exponents

$$\boxed{142.69267, \quad 185.53659, \quad 233.84399}. \quad (52)$$

Thus even this deliberately uniform proof has a unit-leading ledger that lies nominally below the 143, 207, 272 references at all three levels. The table does not include κ_{hom} and therefore is not a complete gate-count comparison.

Proof. The $\ell = 2$ rows are [Theorem 9.2](#); the $\ell = 4$ rows are [Corollary 8.2](#). Both routes use the same accepted-root theorem, exact separator ledger, and complete public filtering. The displayed worst values are the levelwise maxima. $\square$

Theorem 11.2 (Optimized adaptive attack on every published row). *An exact public determinant preflight can replace the direct square by a smaller core without introducing a preferred-core premise: failure invokes the direct complete square. The resulting all-nine table is*

Level	parameters	adaptive selected-degree-free route	ledger exponent	nominal gap
I	(28, 5, 4, 4)	channel core, then ordinary square	128.332	14.668
I	(48, 16, 2, 2)	one eigenblock, then two-block square	132.188	10.812
I	(28, 4, 4, 5)	channel core, then ordinary square	128.332	14.668
III	(40, 7, 4, 4)	channel core, then ordinary square	171.702	35.298
III	(72, 24, 2, 2)	complete two-block square	183.795	23.205
III	(38, 5, 4, 5)	channel core, then ordinary square	171.702	35.298
V	(50, 9, 4, 4)	channel core, then ordinary square	214.644	57.356
V	(96, 32, 2, 2)	complete two-block square	233.844	38.156
V	(52, 6, 4, 6)	channel core, then ordinary square	214.643	57.357

Taking the worst row at each level gives

$$\boxed{132.18807, \quad 183.79463, \quad 233.84399}. \quad (53)$$

The corresponding nominal headroom is

$$\boxed{10.81193, \quad 23.20537, \quad 38.15601} \text{ bits.}$$

At the sensitivity point $\kappa_{\text{hom}} = 2^7$, the remaining headroom is 3.81193, 16.20537, and 31.15601 bits. This is a break-even sensitivity, not a measurement of κ_{hom} .

Proof. Use the exact branch rule of [Theorem 6.8](#). For $\ell = 2$, the Level-I fast branch is the certified one-eigenblock core and the fallback is the complete two-block square; the complete square is already faster at Levels III and V. For $\ell = 4$, use [Lemma 10.2](#) and fall back to [Corollary 8.2](#). The machine-readable ledger mixes branch work before taking logarithms, charges the union of the relevant spectrum tails, and finally normalizes by the exact structural-preflight density. The lower-work sixteen-orbit fallback is reported separately and does not change the levelwise maxima. $\square$

Lower-work and lower-output refinements. Replacing the direct $\ell = 4$ fallback by the sixteen-orbit fallback lowers the six $\ell = 4$ rows to 126.405, 167.289, and 207.844 on the worse shape but does not change any levelwise worst value in (53). Conversely, deliberately using the one-eigenblock $\ell = 2$ core whenever its certificate passes yields a smaller-output fixed-core-free frontier with worst exponents

$$\boxed{132.18807, \quad 185.43454, \quad 237.78324}.$$

Its fast branch is taken with probability above 0.997; the complete square remains an exact fallback.

Scope of the arithmetic unit. The primary tables charge complete validated multiplication circuits for each pointwise product over the base or tower fields $\mathbb{F}_{19}$, $\mathbb{F}_{19^2}$, and $\mathbb{F}_{19^4}$. Extension-field evaluation/interpolation and basis transforms, together with additions, inversions, factorization, control, filtering, memory traffic, and implementation constants, remain inside κ_{hom} . A narrower variable-product-only normalization is retained only in the artifact and is not used for any theorem headline.

12 Complementary streamed-XL and low-state frontiers

The complete-square homotopy theorem is the logical headline, but it does not supply a low-memory implementation. This section records a complementary branch whose matrices can be streamed with linear vector state. Its correctness is exact once a finite quotient certificate is produced; its numerical exponent retains a production selected-degree premise. The two branches therefore answer different questions rather than competing for one headline.

12.1 Four-block special XL

For an A -linear $\ell = 4$ slice of A -dimension s , scalar extension gives four blocks of s variables. Each unordered pair of blocks carries m_1 independent quadratic equations. After adding one homogenizer per block, the Hilbert series used by the published special-XL model is

$$H_{m_1, s}(\mathbf{t}) = \frac{\prod_{a=1}^4 (1 - t_a^2)^{m_1} \prod_{1 \leq a < b \leq 4} (1 - t_a t_b)^{m_1}}{\prod_{a=1}^4 (1 - t_a)^{s+1}}. \quad (54)$$

For a multidegree $\mathbf{d} = (d_1, \dots, d_4)$, the monomial-column count is

$$M_s(\mathbf{d}) = \prod_{a=1}^4 \binom{s + d_a}{d_a}. \quad (55)$$

A negative coefficient of (54) is the selected-degree trigger in the cited model [9]. It predicts that a planted instance has the finite quotient needed for extraction; the theorem below makes the operator and state accounting exact but does not prove that production instances reach such a quotient at that degree.

The exact row count is

$$\begin{aligned} R_s(\mathbf{d}) = & m_1 \sum_{a=1}^4 \binom{s + d_a - 2}{d_a - 2} \prod_{c \neq a} \binom{s + d_c}{d_c} \\ & + m_1 \sum_{1 \leq a < b \leq 4} \binom{s + d_a - 1}{d_a - 1} \binom{s + d_b - 1}{d_b - 1} \prod_{c \notin \{a, b\}} \binom{s + d_c}{d_c}, \end{aligned} \quad (56)$$

where a binomial coefficient with a negative lower argument is zero.

Proposition 12.1 (Streaming the special-XL operator). *Let $A_{\mathbf{d}}$ be the Macaulay matrix at multi-degree $\mathbf{d}$, with $M = M_s(\mathbf{d})$ columns and $R = R_s(\mathbf{d})$ rows. Every row and transpose-row product can be generated from the public quadratic coefficients using $O(s^2)$ scratch space.*

Over an extension $E/\mathbb{F}_{19^4}$, choose a random diagonal $\Delta \in E^{R \times R}$ and put

$$B = A_{\mathbf{d}}^{\top} \Delta A_{\mathbf{d}}.$$

Then

$$\Pr[\ker B \neq \ker A_d] \leq M/|E|.$$

A product by B is evaluated by one pass over the streamed rows and uses $O(M)$ resident field-vector state. Scalar Wiedemann over $E = \mathbb{F}_{19^8}$ requires fewer than

$$16R(s+1)^2M \tag{57}$$

signed $\mathbb{F}_{19^4}$ multiply-accumulates per trial. A conservative cubic preconditioner over $\mathbb{F}_{19^{12}}$ gives the Level-I bound $32R(s+1)^2M$.

Proof. The two sums in (56) count multipliers of the diagonal and off-diagonal equation families. A homogenized quadratic row has at most $(s+1)^2$ terms, so rows and transpose products are generated on demand.

Factor $A_d = UV$ at rank r . If $U^\top \Delta U$ is nonsingular, then $\ker(A_d^\top \Delta A_d) = \ker V = \ker A_d$. By Cauchy–Binet, $\det(U^\top \Delta U)$ is a nonzero multilinear polynomial of total degree r in the diagonal entries. Schwartz–Zippel gives failure at most $r/|E| \leq M/|E|$.

Streaming first computes the row inner product and then accumulates the scaled row. In the quadratic extension, the sparse dot product and accumulation use at most four base-extension products per row term and the diagonal multiplication uses three more. Fewer than $3M$ normal-operator products generate the scalar Wiedemann sequence and reconstruct one null vector. Rounding the resulting constant gives (57); the cubic-extension calculation is analogous. $\square$

12.2 Certified extraction beyond exact corank one

A one-dimensional Macaulay kernel is convenient but is not required for soundness.

Theorem 12.2 (Recovery from a finite commuting border quotient). *Fix a target and slice over a finite field Ω , and let $I \subseteq \Omega[x_1, \dots, x_h]$ be the ideal of the residual equations. Let $\mathcal{B}$ be a finite divisor-closed monomial set containing 1. Suppose the Macaulay row space supplies, for every border monomial $\beta \in \partial\mathcal{B}$, a certified relation*

$$g_\beta = \beta - \sum_{b \in \mathcal{B}} c_{\beta,b} b \in I.$$

Construct the formal multiplication matrices $M_1, \dots, M_h$ on $\Omega\mathcal{B}$. If they commute, then the border relations generate a zero-dimensional ideal $J \subseteq I$, $\mathcal{B}$ is a basis of $\Omega[x]/J$, and every root of I occurs among the roots enumerated from the multiplication tables. FGLM conversion costs $O(h|\mathcal{B}|^3)$ field operations once the tables are available. Complete residual and public verification removes every extra point.

Proof. Commutativity makes $x_i \mapsto M_i$ an algebra homomorphism. Divisor closure identifies each basis monomial with its coordinate vector, and every border relation acts as zero. The border-division theorem therefore gives unique normal forms in $\text{span}_\Omega \mathcal{B}$, so the relations form a border basis of a zero-dimensional ideal J . Because every relation is a Macaulay row combination from I , we have $J \subseteq I$ and hence $V(I) \subseteq V(J)$. Standard FGLM conversion and finite-field factorization enumerate the geometric points of J [11, 12]. $\square$

A quotient of dimension larger than one is therefore a correctness fallback, not a free extension of the scalar-Wiedemann exponent. Obtaining all border relations, block-nullspace work, and resident multiplication-table state must fit the displayed ledger or be charged separately.

12.3 Three-plus-one completion

The fourth eigenblock can be completed linearly after solving the first three. Let E_{ab} be the m_1 equations supported on blocks a, b .

Theorem 12.3 (Exact three-plus-one completion). *Suppose a certified first-stage solver returns a finite set containing every root of the six families E_{ab} with $1 \leq a \leq b \leq 3$. For each partial root $p = (x_1, x_2, x_3)$, substitute into E_{14}, E_{24}, E_{34} and obtain*

$$A_p x_4 = b_p, \quad A_p \in \Omega^{3m_1 \times s}.$$

Discard inconsistent systems. If $\text{rank } A_p = s$, solve for the unique fourth block and retain it exactly when $E_{44} = 0$. If a consistent system has rank below s , declare the trial incomplete unless a separately priced procedure enumerates all completions. Whenever no incomplete case occurs, the retained set contains every root of the complete four-block system. Final descent and verification make the procedure Las Vegas.

Proof. Any complete root restricts to a first-stage root. At that partial root its true fourth block satisfies the displayed linear system. On a trial that declares no incomplete case, every consistent system encountered has full column rank. Hence the true fourth block is the unique completion, and the remaining diagonal family checks it. Conversely every retained point satisfies all ten families. $\square$

12.4 Robust low-state points

The following rows charge the exact streamed row count, homogenizers, preconditioner restarts, rejection density, and a conservative solver certificate rate of at least $1/8$. They retain the production selected-degree premise just described.

Table 4: Complementary robust low-state XL profiles. Work is a base-two AXN arithmetic exponent in the older signed multiply–accumulate ledger; state is an eight-array packed envelope.

Level	Profile	Multidegree	Work	State
I	four blocks, $(s, \tau) = (10, 10)$	$(3, 3, 3, 4)$	141.29	1.53 TiB
III	four blocks, $(s, \tau) = (11, 26)$	$(1, 2, 3, 5)$	200.49	55.4 GiB
V	three-plus-one, $(s, \tau) = (12, 42)$	$(2, 3, 5)$	263.40	9.55 GiB

These points show that the symmetry quotient does not lead only to enormous dense homotopy outputs. They do not replace the selected-degree-free theorem: their unresolved input is whether production systems supply the required finite quotient at the selected multidegree, within the stated block/nullspace budget.

13 Implementation, evidence, and reproducibility

The companion artifact separates theorem checks from experimental evidence. The primary selected-degree-free conclusions no longer rely on sampled production ranks or small-profile Macaulay behavior. Their machine-checkable inputs are the exact public dimensions, the random-XOF entropy calculation, the multihomogeneous coefficient formulas, and the explicit arithmetic circuits.

Table 5: Evidence hierarchy used in this revision.

Check	What is computed	What it establishes
Exact public preflight	Quotient, affine, projection, skew, and determinant ranks for a concrete key	Validity of the exact reduction and the selected solver branch on that key.
Random-XOF theorem	Maximum-atom and fresh-coefficient rank calculations	Density of keys satisfying the aggregate spectrum and structural events in the idealized expansion.
Fixed-key certificate	Projective rank histogram and determinant transcript	Exact replacement for the random-XOF spectrum and fast-core density statements on one key.
Homotopy ledger	Exact Bézout numbers, companion coefficients, field thresholds, and straight-line-program counts	Unit-leading arithmetic theorem; not a wall-clock or low-memory implementation.
Circuit certificate	Exhaustive scalar truth table and exact tower identities	Correctness and size of the charged multiplication primitives.
Streamed XL	Exact Hilbert coefficients, rows, columns, and operator products	State and arithmetic frontier inside the stated selected-degree model.

13.1 Exact checks in the release

The release contains a machine-readable all-nine ledger and an independent checker that verifies:

- (i) all $19^2 = 361$ canonical input pairs of the 150-gate $\mathbb{F}_{19}$ multiplier;
- (ii) the $\mathbb{F}_{19^2}$ and $\mathbb{F}_{19^4}$ tower identities, basis determinant, and gate recurrences;
- (iii) the exact official-target rejection-sampling probability across all nine parameter rows;
- (iv) every 2^{-128} excess-spectrum threshold and the corresponding failure exponent;
- (v) the conservative B^2 separating-form extension optima;
- (vi) the complete-square dimension inequalities for every public row;
- (vii) the projective $\ell = 4$ Jacobian-preflight bounds and the exact $\ell = 2$ sequential product;
- (viii) every adaptive branch mixture and all headline exponents;
- (ix) the optional $55 \rightarrow 16$ Frobenius-orbit count; and
- (x) both structural repair-floor tables.

The public leading-Jacobian preflight is deterministic on a fixed key. The random-XOF calculation is used only to normalize expected work over generated keys; a failed preflight invokes a complete-core fallback.

13.2 What is not claimed

The release does not contain a production-size end-to-end forgery or a measured symbolic-homotopy implementation. In particular, the paper does not claim that κ_{hom} is small, that the direct parametrization fits in commodity memory, or that the unit-leading exponent is a complete NIST gate count. The AXN basis prices the multiplication schedule selected by the arithmetic theorem. Control, data movement, indexing, factorization, communication, and peak state remain outside that leading schedule and inside κ_{hom} .

The low-state XL branch has the opposite profile: its streamed operator and state are explicit, but its production selected degree is not proved. A production border-basis transcript or a full-size solve would settle that remaining branch-specific question without changing the exact quotient or the selected-degree-free theorem.

13.3 Fixed-key end-to-end certificate format

A deterministic record for one public key consists of:

1. the public relation, output split, and all exact rank preflights;
2. the selected slice and either its exact projective rank histogram or a stronger unit-ideal certificate;
3. the fast-core leading-Jacobian witness, if the fast branch is taken;
4. the homotopy random coins, parametrization, and all candidate filters;
5. the reconstructed signature and the unmodified verifier transcript.

Such a record would remove the random-XOF key-density statement for that key and directly measure κ_{hom} and resident state.

14 A combined structural repair barrier

The feature-count floor and the fresh-square floor are independent necessary conditions and should be combined.

Proposition 14.1 (Universal fresh-square dimension bound). *Let $A = \mathbb{F}_{q^d}$ and retain the public vinegar summand V' of base-field dimension $d(v-1)$. Suppose an exact affine reduction has M' outputs, K' quotient quadratics, and full affine rank $M' - K'$. Then*

$$\dim_{\mathbb{F}_q}(V' \cap \ker C_{R,v}) \geq d(v-1) - (M' - K').$$

If $M' \leq d(v-1)$, dimensions alone leave a complete ordinary K' -dimensional square slice. Eliminating this guarantee requires $M' > d(v-1)$. Under the coupling $M' = dx'$, this forces $x' \geq v$ and $M' \geq dv$.

Proof. Use

$$\dim(V' \cap \ker C_{R,v}) \geq \dim V' - \text{rank } C_{R,v} = d(v-1) - (M' - K').$$

The right side is at least K' exactly when $M' \leq d(v-1)$. Negate and round to the next multiple of d . $\square$

Let $K_{\text{ord}} = \min\{M, m_1 d^2\}$ be the old ordered-label cap. Retaining symmetric forms requires $m'_1 \geq \lceil K_{\text{ord}} / \binom{d+1}{2} \rceil$ and $M' \geq K_{\text{ord}}$. Under $M' = dx'$ and $m'_1 = \lceil x'/d \rceil$, all three necessities combine as follows.

Theorem 14.2 (Combined necessary output floor). *Any redesign that retains symmetric public forms, the public vinegar count, the public-vinegar affine geometry above, full affine rank $M' - K'$, and the Version 2.3 output/base-form coupling, while both restoring K_{ord} and eliminating the guaranteed fresh square slice by dimensions, must satisfy*

$$x' \geq \max \left\{ v, \left\lceil \frac{K_{\text{ord}}}{d} \right\rceil, d \left(\left\lceil \frac{K_{\text{ord}}}{\binom{d+1}{2}} \right\rceil - 1 \right) + 1 \right\}, \quad M' = dx'. \quad (58)$$

For the nine official rows:

Level	parameters	current M	necessary M'	increase
I	(28, 5, 4, 4)	80	116	45.00%
I	(48, 16, 2, 2)	64	96	50.00%
I	(28, 4, 4, 5)	80	116	45.00%
III	(40, 7, 4, 4)	112	180	60.71%
III	(72, 24, 2, 2)	96	144	50.00%
III	(38, 5, 4, 5)	100	152	52.00%
V	(50, 9, 4, 4)	144	228	58.33%
V	(96, 32, 2, 2)	128	192	50.00%
V	(52, 6, 4, 6)	144	228	58.33%

Thus every parameter-only repair retaining these hypotheses needs at least 45.00% more verifier outputs, and the largest necessary increase is 60.71%, before adding any security margin. These are necessary, not sufficient, repaired parameters.

Proof. The second and third terms in (58) are the two ordered-cap requirements under $m'_1 = \lceil x'/d \rceil$. The term v is Proposition 14.1. Their maximum is therefore necessary. Substitution gives the table. $\square$

14.1 A stronger chosen-message square-persistence barrier

The preceding floor assumes a full-rank affine lift and asks dimensions to remove a square slice from every target fiber. A zero-offset attacker has a second option: give up affine consistency for all targets, filter chosen-message targets through the quadratic image, and recover the entire common-column domain. Eliminating that fallback by dimensions alone is much more expensive.

Proposition 14.3 (Zero-offset chosen-message square persistence). *Consider any symmetric redesign for which the zero-offset common-column relation with $R_0 = 1$ is defined. Let Q_R be its quadratic output matrix and put $\rho = \text{rank } Q_R$. Public row reduction gives an invertible output change under which the zero-offset verifier is equivalent to*

$$g_\rho(u) = z \in \mathbb{F}_q^\rho, \quad t_{\text{con}} = 0 \in \mathbb{F}_q^{M'-\rho}, \quad (59)$$

where g_ρ consists of ρ independent quadratic output combinations. For an exactly uniform target, the consistency event in (59) has probability $q^{-(M'-\rho)}$ and, conditioned on it, z is uniform in $\mathbb{F}_q^\rho$.

If the public vinegar summand V' has dimension $d(v-1)$ and

$$\rho \leq d(v-1), \quad (60)$$

then dimensions alone leave an ordinary ρ -dimensional slice $L \subseteq V'$ on which the complete residual system is square: it has ρ equations in ρ variables. Consequently, ruling out this chosen-message square by dimensions alone requires

$$\rho > d(v-1). \quad (61)$$

Proof. With zero offsets, the substituted verifier is $Q_R z_{\text{sym}}(u) = t$ and contains no affine or constant term. Choose a row basis of Q_R and extend it to an invertible basis change on the M' output coordinates. The first ρ transformed rows are the independent residual quadratics g_ρ ; the remaining rows vanish identically on the quadratic image and therefore impose the consistency equations in (59). An invertible linear transformation sends a uniform target to independent uniform residual and consistency coordinates, proving the distribution statement. Under (60), choose any ρ -dimensional linear subspace of V' and restrict all ρ residual equations to an affine coset of it. This gives the asserted complete square system. $\square$

Symmetry bounds the actual quotient rank by

$$\rho \leq \min \left\{ M', m'_1 \binom{d+1}{2} \right\}. \quad (62)$$

Hence making (61) even possible forces both terms on the right of (62) above $d(v-1)$.

Theorem 14.4 (Dimension-only floor against the zero-offset fallback). *Retain symmetry, the public vinegar count, and the Version 2.3 coupling*

$$M' = dx', \quad m'_1 = \left\lceil \frac{x'}{d} \right\rceil.$$

Put

$$C_d = \binom{d+1}{2}, \quad m_{\text{sq}} = \left\lfloor \frac{d(v-1)}{C_d} \right\rfloor + 1.$$

A necessary condition for eliminating the zero-offset chosen-message square by dimensions alone is

$$x' \geq \max \{v, d(m_{\text{sq}} - 1) + 1\}, \quad M' = dx'. \quad (63)$$

If the redesign also restores the old ordered-label cap K_{ord} , the right side of (63) must additionally be maximized with

$$\left\lceil \frac{K_{\text{ord}}}{d} \right\rceil, \quad d \left(\left\lceil \frac{K_{\text{ord}}}{C_d} \right\rceil - 1 \right) + 1.$$

For the nine official rows, the zero-offset condition already dominates the ordered-cap terms:

Level	parameters	current M	dimension-only M'	increase
<i>I</i>	(28, 5, 4, 4)	80	164	105.00%
<i>I</i>	(48, 16, 2, 2)	64	126	96.875%
<i>I</i>	(28, 4, 4, 5)	80	164	105.00%
<i>III</i>	(40, 7, 4, 4)	112	244	117.86%
<i>III</i>	(72, 24, 2, 2)	96	190	97.92%
<i>III</i>	(38, 5, 4, 5)	100	228	128.00%
<i>V</i>	(50, 9, 4, 4)	144	308	113.89%
<i>V</i>	(96, 32, 2, 2)	128	254	98.44%
<i>V</i>	(52, 6, 4, 6)	144	324	125.00%

Thus eliminating this fallback by dimensions alone requires 96.875%–128.00% more outputs across the published rows.

Proof. Condition (61) and the two caps in (62) require $dx' > d(v-1)$ and $m'_1 C_d > d(v-1)$. The first inequality is equivalent to $x' \geq v$. The second requires $m'_1 \geq m_{\text{sq}}$, and the least integer x' with $\lceil x'/d \rceil \geq m_{\text{sq}}$ is $d(m_{\text{sq}} - 1) + 1$. This proves (63). The two extra ordered-cap terms are the same feature and output necessities used in Theorem 14.2. Direct substitution gives the table. $\square$

How to interpret the stronger floor. This is deliberately a dimension-only theorem, not a claim that every secure repair must literally double its outputs. A redesign below the table can instead leave the square system present and try to obtain security from the chosen-message consistency factor $q^{M'-\rho}$, from the cost of solving the square system, or from a changed target interface. It must then publish and price those ingredients explicitly. The theorem gives a sharp trichotomy: break the symmetric/affine architecture, nearly double the outputs to remove the fallback by dimensions, or retain the fallback and justify its complete chosen-message cost.

15 Conclusion

The $q = 19$ Version 2.3 construction loses quadratic directions for every symmetric public key. Ordered power-pair labels are not ordered polynomials: swapping the labels leaves the scalar quadratic unchanged before the public outer map. Exact output splitting turns that identity into complete forgery systems without discarding a verifier equation, and the block-symmetry rejection rule does not remove them.

The main theorem is now simple. Every published row leaves a complete square residual system, and solving that system directly gives selected-degree-free, fixed-core-free Las Vegas recovery routes whose unit-leading ledgers lie nominally below the numerical references at all three security levels. This proof uses no selected Macaulay degree, exact corank, unique-root assumption, chosen Jacobian core, or orbit enumeration. Exact public determinant tests provide faster adaptive routes, but they are optimizations rather than premises: failure invokes a complete-core fallback.

With the audited conservative separator and Jacobian-density bounds, the direct all-nine worst exponents are 142.693, 185.537, and 233.844. The adaptive values are 132.188, 183.795, and 233.844. The explicit multiplier κ_{hom} remains, so these are operation-count and break-even theorems rather than production timings or complete attack circuits. The complementary streamed-XL branch supplies an explicit low-state frontier under its separate production selected-degree premise.

A repair retaining the architecture cannot be a small rejection patch. Restoring the former feature cap and eliminating the guaranteed full-affine square already requires 45.00%–60.71% more verifier outputs under the current coupling. Eliminating the zero-offset chosen-message square by dimensions alone requires 96.875%–128.00% more. A smaller redesign must retain and explicitly price the square system and target filter, change the target interface, or abandon the symmetric/affine architecture.

A Exact fixed-key spectrum certificates

The random-XOF theorem is not required for a statement about a concrete key. This appendix gives a finite exact certificate for the aggregate spectrum.

Choose a basis $y_0, \dots, y_{h-1}$ of a fixed slice L and write

$$T(Y) = \sum_{i=0}^{h-1} Y_i T_{y_i} \in \mathbb{F}_q[Y_0, \dots, Y_{h-1}]^{K \times N}.$$

For $0 \leq j < h$, use the disjoint normalized projective chart $Y_0 = \dots = Y_{j-1} = 0$, $Y_j = 1$, and put $R_j = \mathbb{F}_q[Y_{j+1}, \dots, Y_{h-1}]$. Let $T^{(j)}$ be the resulting matrix. For $t \geq 0$, define

$$I_{j,t} = \left\langle Y_i^q - Y_i \ (i > j), \quad \text{all } (t+1) \times (t+1) \text{ minors of } T^{(j)} \right\rangle \subseteq R_j. \quad (64)$$

Put $n_{j,\leq t} = \dim_{\mathbb{F}_q}(R_j/I_{j,t})$, $n_{j,\leq -1} = 0$, and

$$N_t = \sum_{j=0}^{h-1} (n_{j,\leq t} - n_{j,\leq t-1}). \quad (65)$$

Proposition A.1 (Projective rank-histogram certificate). *The integer N_t is exactly the number of projective directions $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$ for which $\text{rank } T_y = t$. Consequently*

$$\bar{\eta}_L^U = (q-1) \sum_t N_t q^{-t}. \quad (66)$$

Reduced Gröbner bases with their standard monomial sets, or polynomial reduction transcripts proving the unit-ideal and quotient-dimension claims, form independently checkable fixed-key certificates.

Proof. The normalized charts contain exactly one representative of every projective $\mathbb{F}_q$ -direction. The field equations identify

$$R_j / \langle Y_i^q - Y_i : i > j \rangle \cong \prod_{a \in \mathbb{F}_q^{h-j-1}} \mathbb{F}_q$$

by evaluation. Every ideal in this product of fields is radical, and its quotient dimension is the number of coordinates on which all generators vanish. The minors in (64) vanish exactly when $\text{rank } T^{(j)}(a) \leq t$. Thus $n_{j, \leq t}$ counts rank-at-most- t points in chart j , and differencing gives (65). Every projective direction has $q - 1$ nonzero scalar representatives and rank is invariant under base-field scaling, proving (66). $\square$

A stronger certificate that $\text{rank } T_y \geq r + 1$ for every nonzero y consists of unit-ideal transcripts for all $I_{j,r}$. The complete histogram is more flexible: a small exceptional low-rank locus can be charged exactly instead of controlling the whole pencil by its minimum rank.

B Dimension formulas and exact numerical rows

For a Version 2.3 row,

$$n = v + o, \quad m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad M = or\ell, \quad N_0 = n\ell, \quad K = m_1 \binom{\ell + 1}{2}.$$

For a full affine lift with rank $C_{R,v} = M - K$,

$$\dim \ker C_{R,v} = N_0 - M + K.$$

For a structured lift whose complete offset orbit has rank $m_1 \ell^2$,

$$\dim_A H_{R,v}^F = n - m_1 \ell.$$

The direct ordinary fresh-slice lower bound for $\ell = 4$ is

$$\dim(V' \cap \ker C_{R,v}) \geq 4(v - 1) - (M - K).$$

Table 6: Audited exact unit-leading exponents underlying the rounded main text. “Direct” solves the complete square; “adaptive” uses an exact fast-core preflight and a complete-core fallback.

Level	Parameters	Direct	Adaptive	Nominal adaptive gap
I	(28, 5, 4, 4)	142.692668	128.331985	14.668015
I	(48, 16, 2, 2)	132.970670	132.188072	10.811928
I	(28, 4, 4, 5)	142.692082	128.331455	14.668545
III	(40, 7, 4, 4)	185.536587	171.702386	35.297614
III	(72, 24, 2, 2)	183.794625	183.794625	23.205375
III	(38, 5, 4, 5)	185.535753	171.701565	35.298435
V	(50, 9, 4, 4)	227.646114	214.644029	57.355971
V	(96, 32, 2, 2)	233.843987	233.843987	38.156013
V	(52, 6, 4, 6)	227.645066	214.642984	57.357016

All values charge the 150, 692, and 2628-gate multiplication circuits as appropriate and use the conservative B^2 forbidden-vector separator. For $\ell = 4$, the Jacobian-preflight density uses the projective right-kernel union bound, not an independence claim after a biased linear transform.

C Arithmetic-circuit certificate details

The scalar multiplier consumes two little-endian five-bit canonical integers in $\{0, \dots, 18\}$ and returns the canonical five-bit representation of their product modulo 19. Its 150 gates are two-input AND, XOR, and XNOR gates with free constants, wires, fan-out, and structural sharing. The released netlist is tested on all 361 valid input pairs. Its SHA-256 digest is

e1708a32c1475ad14295c2b34fed976fb09577b127810d29bf237db18aabc131

for the audited balanced-netlist representation.

The tower recurrences are

$$\mathbb{F}_{19^2} = \mathbb{F}_{19}[u]/(u^2 + 1), \quad \mathbb{F}_{19^4} = \mathbb{F}_{19^2}[v]/(v^2 - (1 + u)).$$

Three-product Karatsuba at each level, together with the exact linear transforms stated in [Theorem 7.1](#), gives complete multiplication counts 692 and 2628. The artifact independently verifies the tower identities, the basis determinant, and random products against direct polynomial arithmetic.

These circuits normalize only the pointwise multiplication schedule. They are not lower bounds and do not price the whole solver. Additions, inversions, factorization, fixed extension transforms, filtering, control, indexing, communication, and memory remain in κ_{hom} .

D Reproduction and package contents

The source package is self-contained. Its root contains the complete paper source, bibliography, compiled PDF, build instructions, and hashes. The `artifact/` directory contains:

- the exact all-nine numerical ledger and generator;
- an independent checker for dimensions, spectrum tails, separator fields, adaptive mixtures, orbit count, and repair floors;
- the complete 150-gate scalar multiplier netlist;
- the field-tower construction and verification scripts; and
- the generated circuit and numerical JSON records.

A clean reproduction is:

```
python3 artifact/verify_final_results.py
latexmk -pdf -interaction=nonstopmode -halt-on-error paper.tex
```

The checker is deterministic and does not require network access. The paper build uses only the files in the package. The supplied SHA256SUMS records every distributed file.

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